Data Structure and Algorithm Design

ME/MSc (Computer) – Pokhara University

Prepared by:

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Chapter 2:
Advanced Data Structures and Algorithms (11 hrs)

Outline

1. Advanced Trees
   - B and B+ Trees
2. Advanced Graph
   - Planar Graphs and its applications in Geographic and Circuit Layout Design
3. Network Flow Algorithms
   - Ford-Fulkerson Algorithm
   - Edmonds-Karp Algorithm
4. Advanced Shortest Path Algorithms
   - Bellman-Ford Algorithm
   - Johnson's algorithm
5. Graphical Data Structures
   - Quadtrees, KD Trees, R-Trees
6. Computational Geometry
   - Convex Hull and Voronoi Diagrams
7. Case Studies in Red-Black Tree and Its Applications, Applications of Directed Acyclic Graph, Applications of Minimum Spanning Tree in Network Design

Balance Binary Tree

- What if the balanced binary search tree is very-very big and does not fit in RAM?
  - Need to store in the secondary memory (HDD or SSD)
  - This requires read/write operations on HDD
  - Very very costly
    - Due to this cost of read/write operations on HDD, the balanced binary search tree becomes greatly inefficient when the data need to store are very large and stored in secondary memory.
  - Problems of Secondary Storage
    - RAM is more expensive per bit than HDD or SSD
    - SSDs are more faster than HDD
    - However, HDD have much higher capacities than SSDs
    - HDD are more cheaper than SSDs
    - We cannot replace HDDs by SSDs while storing huge amount of data.
    - So need to consider HDDs as a secondary storage

Problem: Secondary Storage

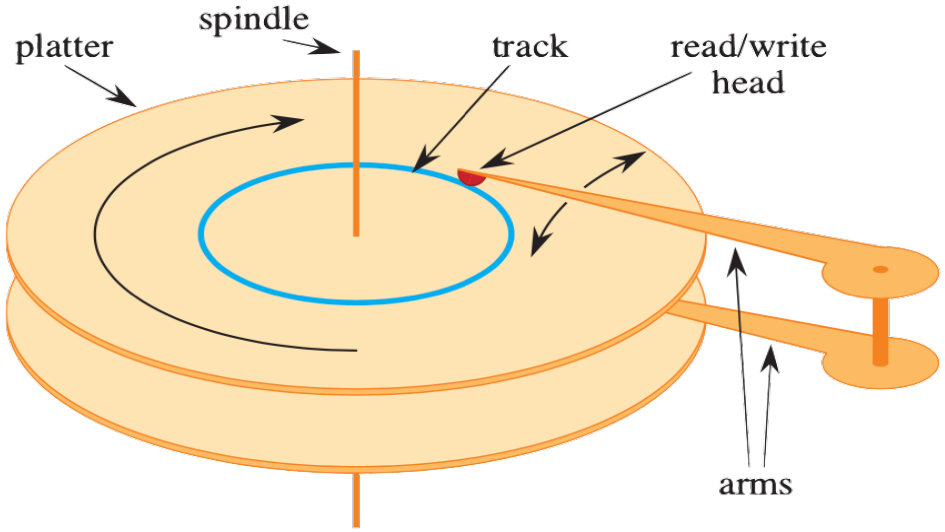
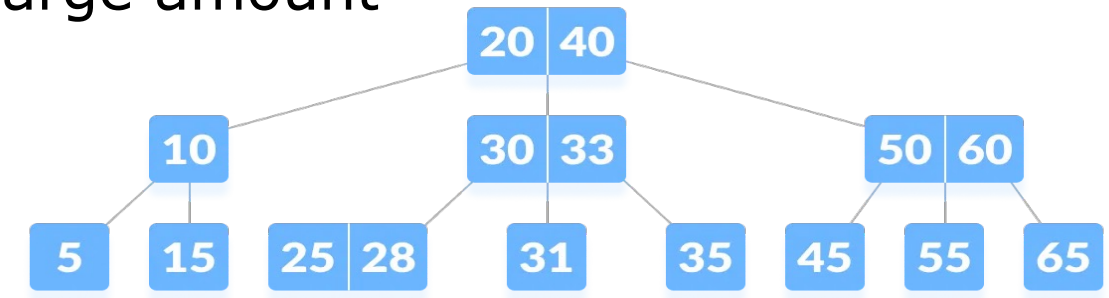


Figure 18.2 A typical magnetic disk drive. It consists of one or more platters covered with a magnetizable material (two platters are shown here) that rotate around a spindle. Each platter is read and written with a head, shown in red, at the end of an arm. Arms rotate around a common pivot axis. A track, drawn in blue, is the surface that passes beneath the read/write head when the head is stationary.

- if a computer waits a full rotation for a particular item to come under the read/write head, it could access main memory more than 1,00,000 times during that span
- The difference in access times for main memory compared with disk drives is enormous
- So to reduce the time spent waiting for mechanical movements (***latency***) disk drives access not just one item but several at a time.
  - Information is divided into a number of equal-sized ***blocks*** of bits (Typical block size: 512 to 4096 bytes)
  - Once the read/write head is positioned correctly and the platter has rotated to the beginning of the desired block, the disk drive can quickly read or write large amounts of data

Problem: Secondary Storage

- Accessing a block of information and reading it from a disk drive (disk access) takes longer than processing all the information read (CPU computing time)
- So, Total Running Time = Time (**# of disks accesses** + **CPU computing**)
- # of disk accesses is measured in terms of # of blocks of information read from or written to the disk drive.
- Can we avoid to store the huge amount of data in secondary memory?
  - No, we have to store the data in secondary memory when it does not fit into RAM.
- Note the another problem of balanced BST
  - Branching factor = 2 at most
  - This causes
    - a very large depth of BST when is very large
    - Increase in # of disk accesses when the BST is stored in HDD
- ***Solution***: Decrease the depth of BST by large amount
  - Increase branching factors
  - Keep multiple keys in a single node
  - Example: B-tree



Indexing

- Indexing mechanisms are used to speedup access to desired data.
- Search Key-attribute to set of attributes used to lookup records in a file.
- An index file consists of records (called index entries) of the form

Search Key	Pointer

- Index files are typically much smaller than the original file
- Two basic kinds of indices:
  1. Ordered indices: search keys are stored in sorted order
  2. Hash indices: search keys are distributed uniformly across "buckets" using a "hash function".
- Index Evaluation Metrics:

  1.Access time 2. Insertion time 3. Deletion time 4. Space overhead

  5.Access types supported efficiently. E.g.,
  - records with a specified value in the attribute
  - or records with an attribute value falling in a specified range of values.
  - This strongly influences the choice of index, and depends on usage.

Indexing

Ordered Indices

- In an ordered index, index entries are stored sorted on the search key value. E.g., author catalog in library.

1. Primary index: in a sequentially ordered file, the index whose search key specifies the sequential order of the file.
   - Also called clustering index
   - The search key of a primary index is usually but not necessarily the primary key.
2. Secondary index: an index whose search key specifies an order different from the sequential order of the file.
   - Also called non-clustering index.
3. Index-sequential file: ordered sequential file with a primary index.

Indexing

- A **database index** is a data structure that improves the speed of data retrieval operations on a database table at the cost of additional writes and storage space to maintain the index data structure.

Types of Indexes

1. Sparse Index

- One entry per data block
- Identifies the first record of the block
- Requires data to be sorted

Advantages :

- Occupy much less space
- Can keep more of it in main memory
- *Faster access*

2. Dense Index

- One entry per record
- Data do not have to be sorted

Advantages:

- Can tell if a given record exists without accessing the file
- Do not require data to be sorted

Sparse vs Dense index

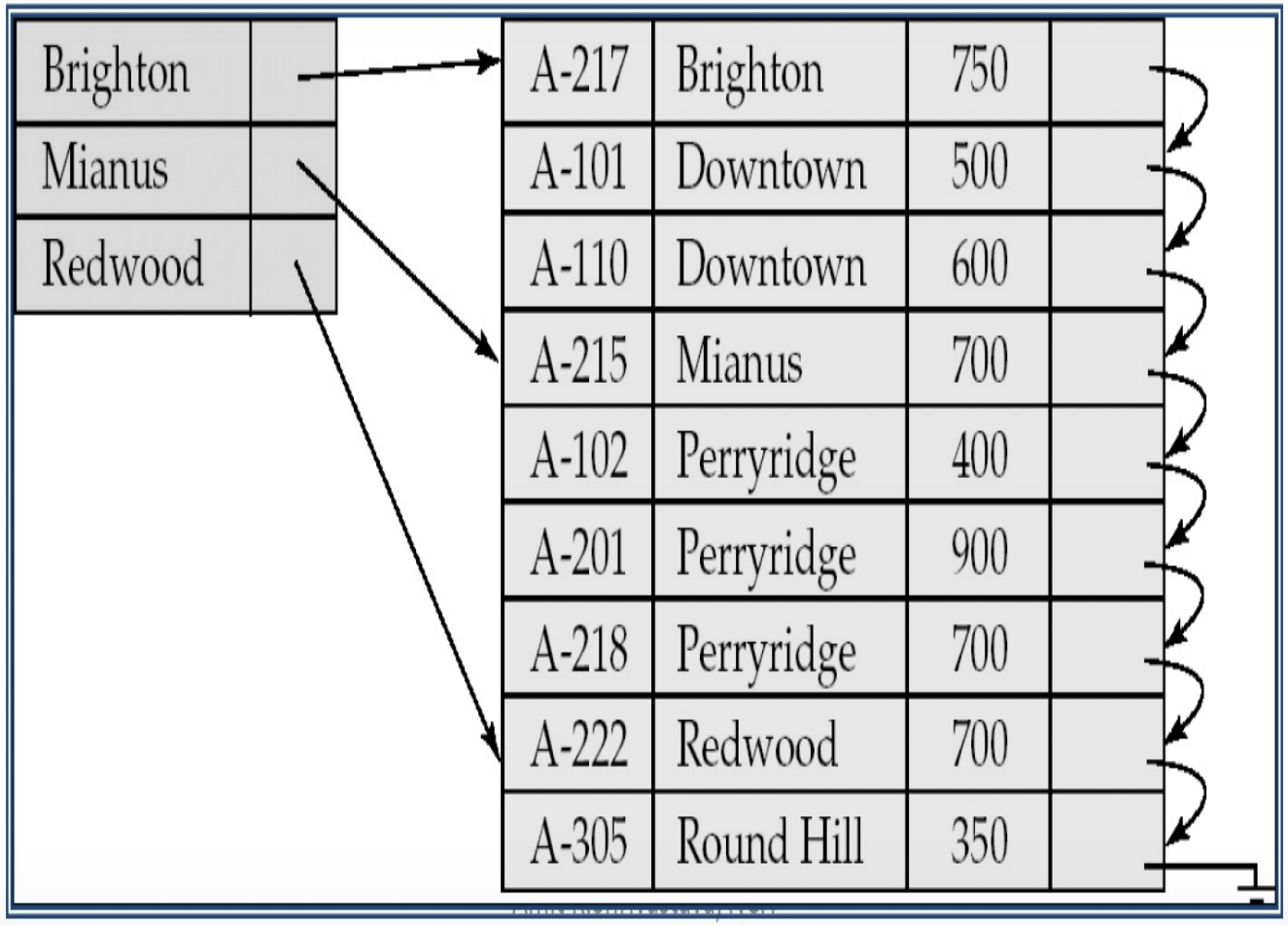


Fig: Sparse index

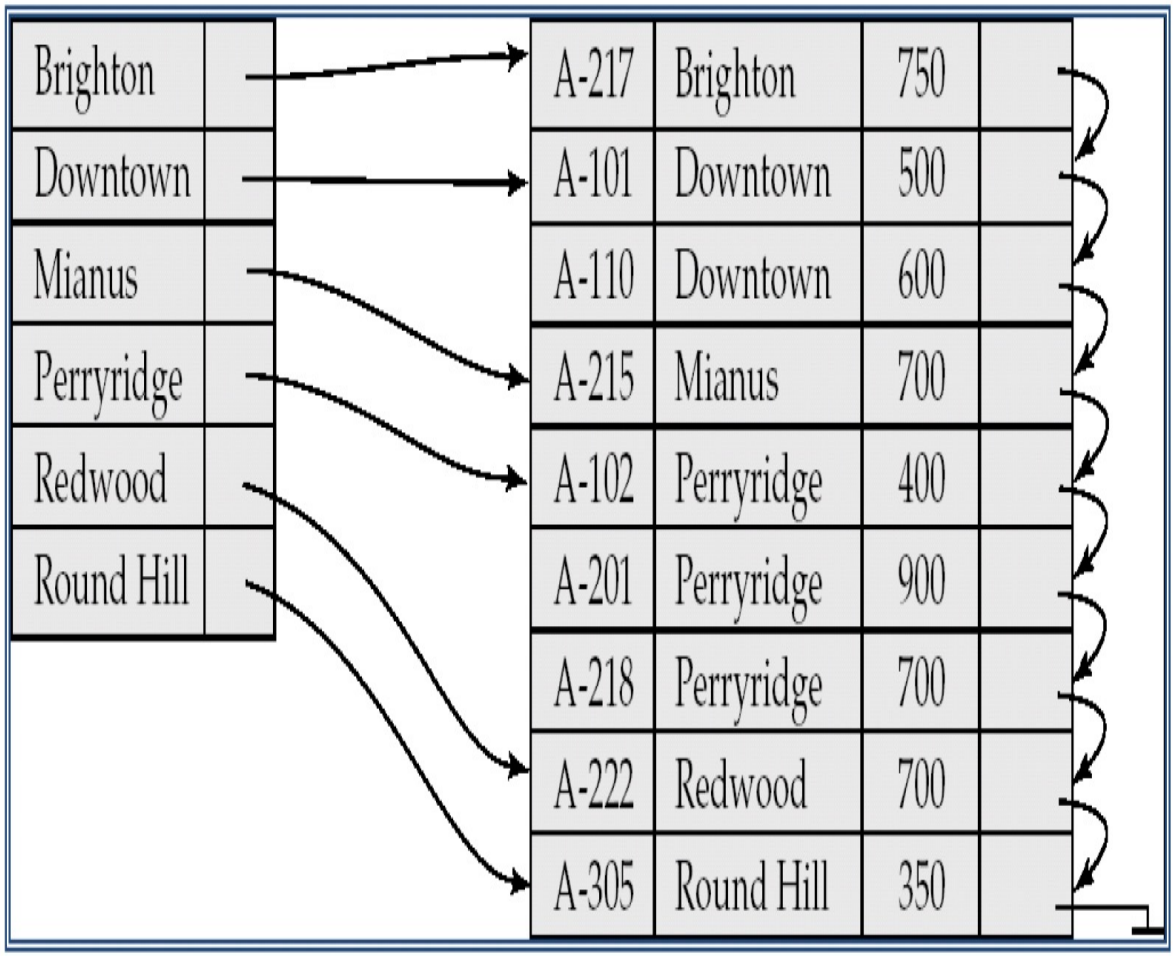


Fig: Dense index

Indexing

Database Indexes:

- Index is a pointer to data in a table. The pointer helps the storage engines to locate data with a reduced latency. Any kind of index usually slows down writes, because the index also needs to be updated every time data is inserted. Therefore, it is an important trade-off in storage systems to identify the required indexes for a dataset.

Disk Indexes:

- Data is ultimately stored in physical disks. Any block on the disk can be accessed via the sector and the track number. Data byte on a particular block can be located with the sector, track and the offset value.
- Example: a **block size of 512 bytes and a row size of 128 bytes**, a total of (512/128), 4 rows can be stored in a block. Now, to store 100 rows, a total of 25 blocks on the disk will be utilized.

Indexing

- Eg: To look for a particular key in the database, the engine needs to access at most 25 blocks. To to reduce access time, we need to index table where each row will have the key and pointer to every value in the database. Considering the size of each row of the index table to be 16 bytes, it would take at most 4 disk blocks.

- To locate a row, the database uses the index table (4 blocks) to find the key, then reads the record(1 block) from the data table, reducing access time from 25 blocks to 5. However, updating the index table is required for every new key insertion.

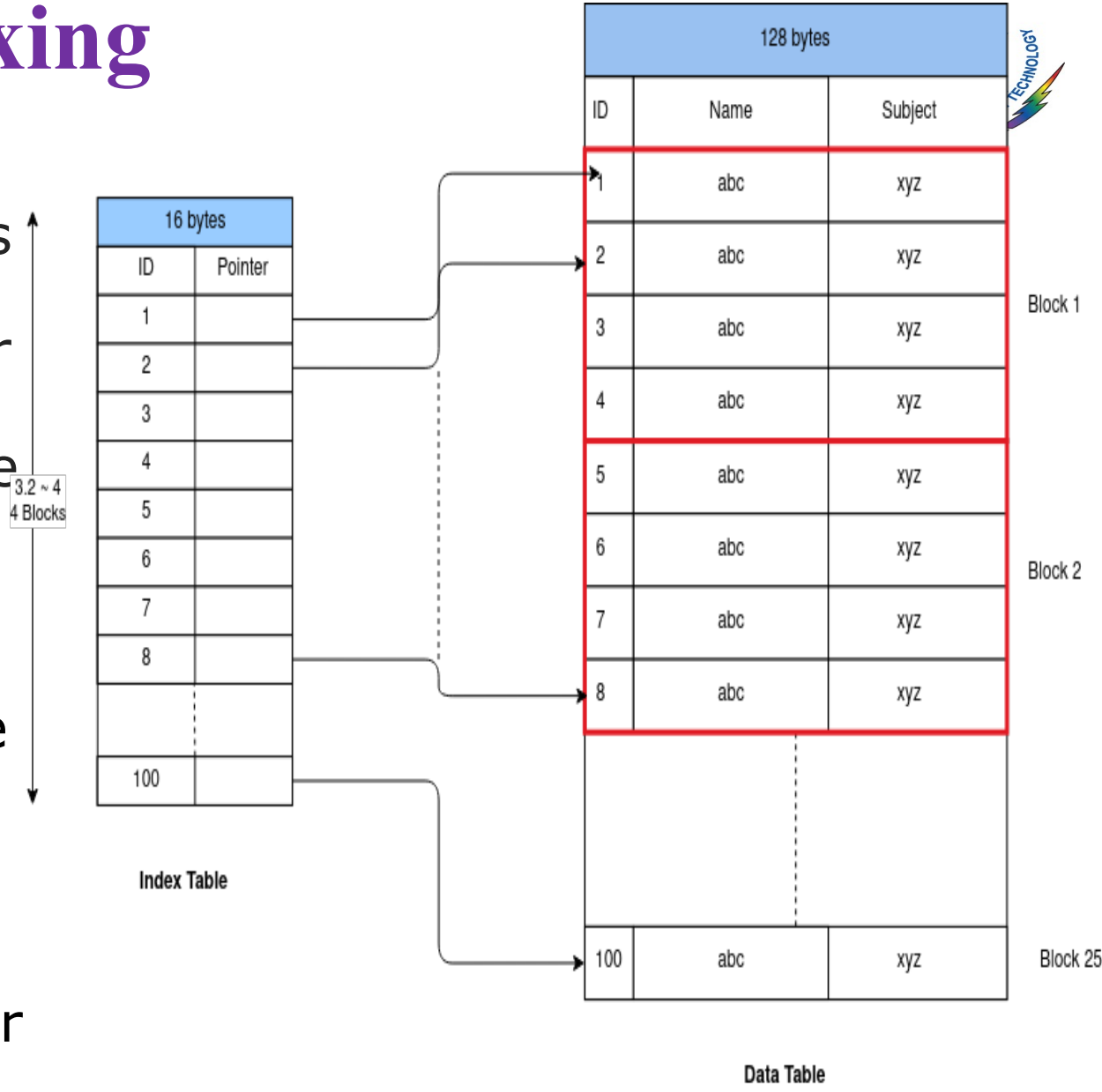


Fig: Example of a primary index

Illustration of Calculation

Table Storage:

Block Size: 512 bytes.

Row Size: 128 bytes.

- **Rows per Block:**
  Rows per block=Block size/Row size=512/128=4 rows per block.
- **Example Data Table:**
- Suppose the table has **100 rows**, each storing a key and associated data.
- To store all these rows:

Number of blocks required=Total rows/Rows per block=100/4=25 blocks.

Block 1: Row 1, Row 2, Row 3, Row 4
Block 2: Row 5, Row 6, Row 7, Row 8
...
Block 25: Row 97, Row 98, Row 99, Row 100

Key Lookup Without Indexing:

To find a particular key (e.g., Key 97), the database engine may need to search through all 25 blocks sequentially.

•This means **25 disk accesses**, which is time-consuming.

- **Index Table Design:**

Row Size in Index Table: 16 bytes (each row contains the key and a pointer to the data block).

Rows per Block in Index Table:
Rows per block=Block size/Row size=512/16=32 rows per block.

Storage for Index Table:

- For the original table of **100 rows**:
- Number of blocks required for the index table=

Total rows/Rows per block=100/32≈4 blocks.
Number of blocks required for the index table=

The index table is distributed over **4 blocks**:

Index Block 1: Index for Rows 1–32
Index Block 2: Index for Rows 33–64
Index Block 3: Index for Rows 65–96
Index Block 4: Index for Rows 97–100

Key Lookup With Indexing:
1.To find Key 97:

- First, access the **index table** (at most 4 blocks) to locate the block containing Key 97.
- The pointer in the index table directly points to **Block 25**, where Key 97 is stored.
- Access Block 25 to retrieve the actual data.

1.Total Disk Accesses:

- Access the **index table**: **4 blocks** (at most).
- Access the **data block**: **1 block** (Block 25).
- **Total disk accesses = 4 + 1 = 5** (significantly fewer than 25).

Indexing

Multi Level Indexing

- With an increase in the number of records (let's say 1000 rows), accessing the index table itself will require more disk blocks to be read. To further reduce this time, another level of indexing can be done. Another level of index can be created that stores the pointer to each block of the first-level index table.
- In example, another index table which stores the block address (*ID 1 to 32 and so on*) of the keys from the first index table.
- The multi-level index table requires only 2 blocks to get accessed, now the record can be accessed in (2 + 1 + 1)= 4 block reads for a 1000 rows data table.
- As the scale of the dataset increases, more levels of indexing can be added

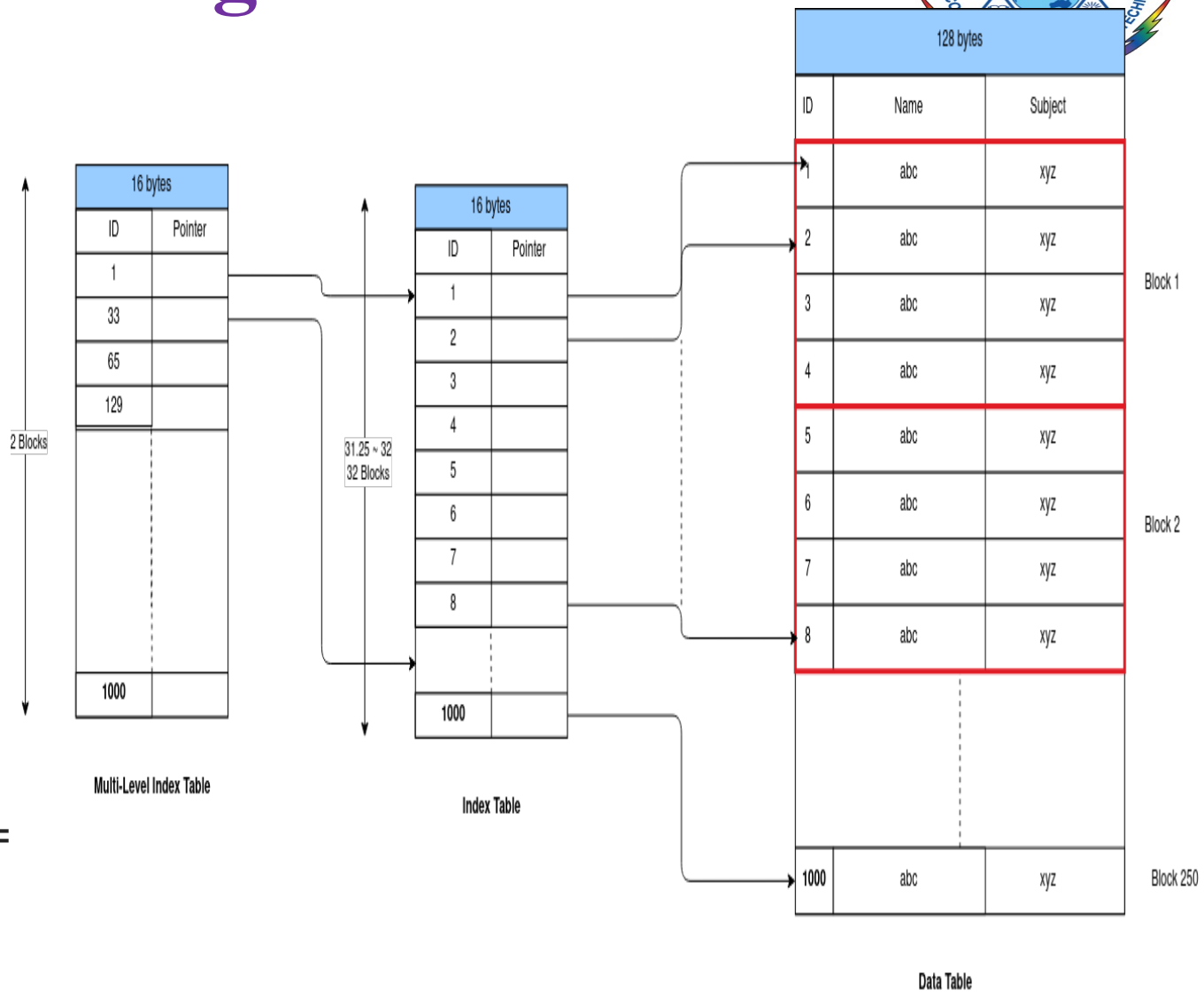
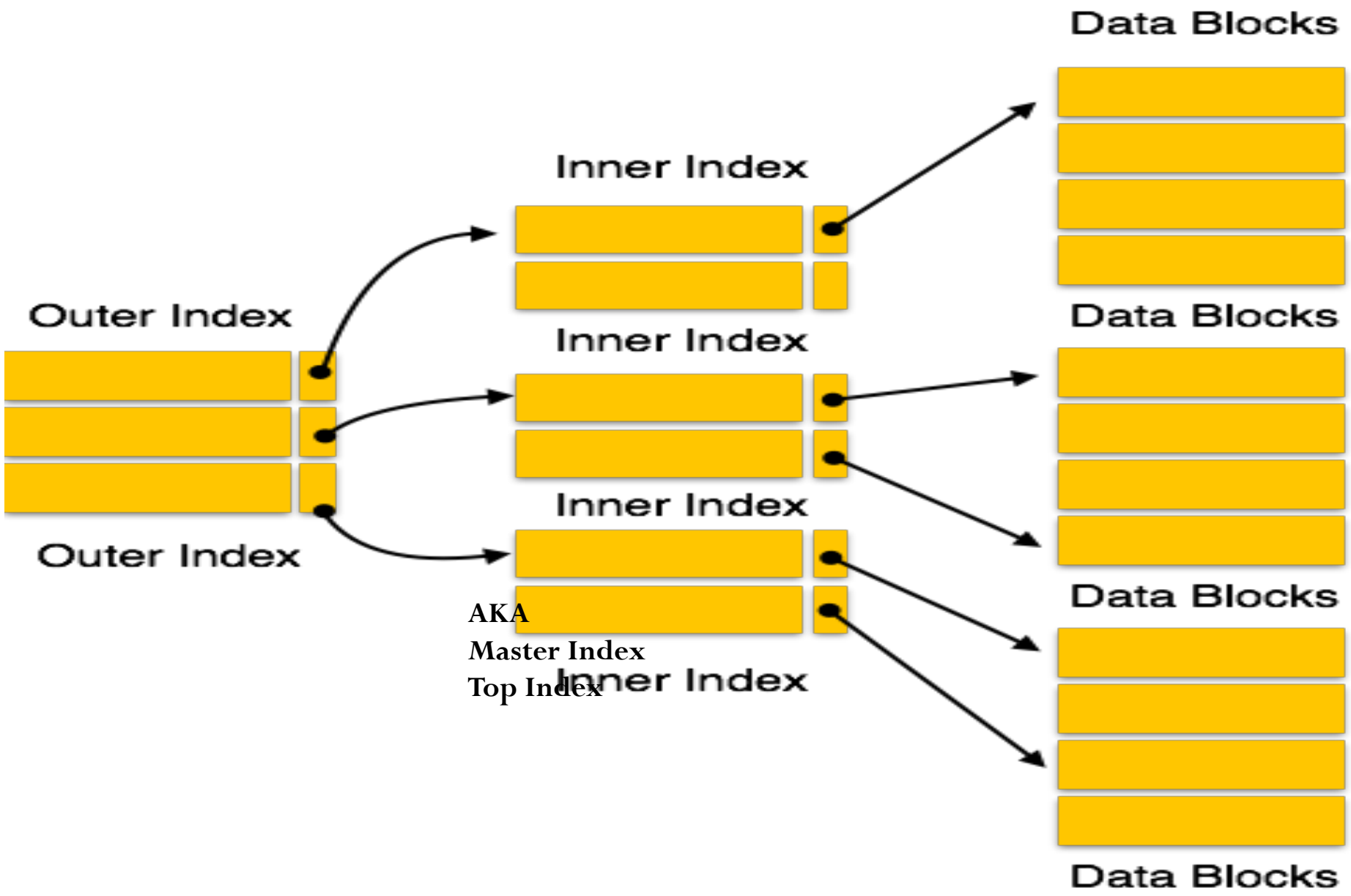


Fig: Multi-Level Indexing

Two levels Index Structure



Advance Tree

M-Way Trees

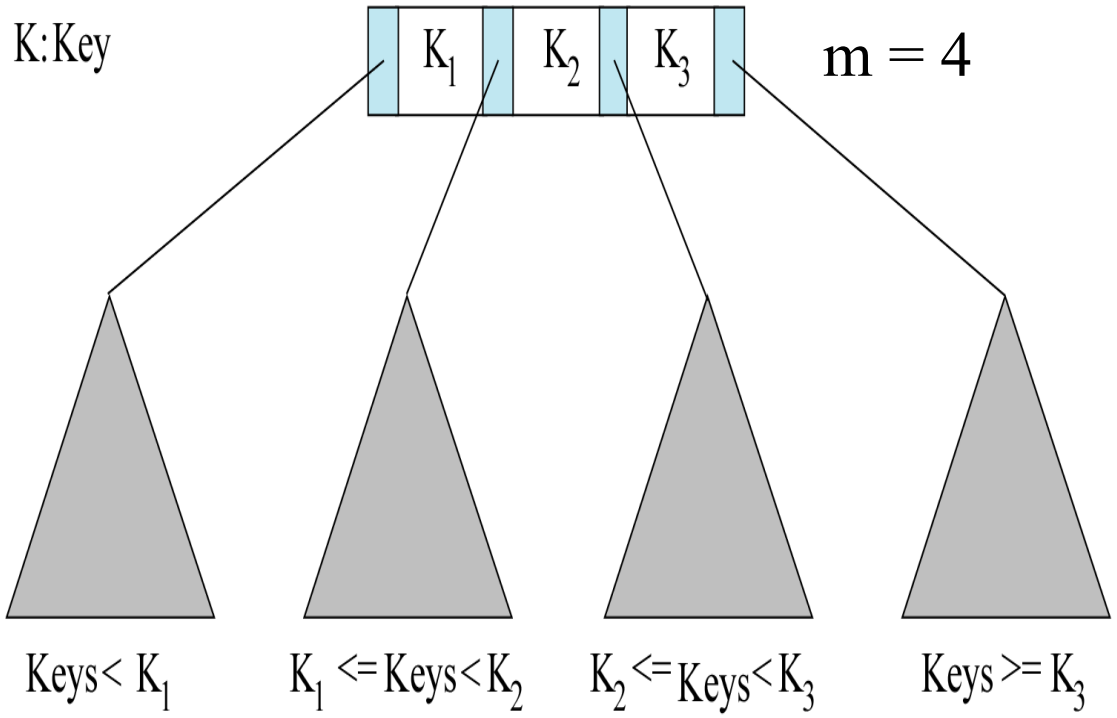
- M-way tree is a search tree.
- Each node of tree can have from zero to **m** subtrees.
- **m** is defined as the order of the tree.
- In an **m-Way** tree of order **m**, each node contains a maximum of **m – 1** elements and **m** children.
- The goal of **m-Way** search tree of height h calls for **O(h)** no. of accesses for an insert/delete/retrieval operation.
- The number of elements in an **m-Way** search tree of height **h** ranges from a minimum of **h** to a maximum of $\mathbf{m^h-1}$

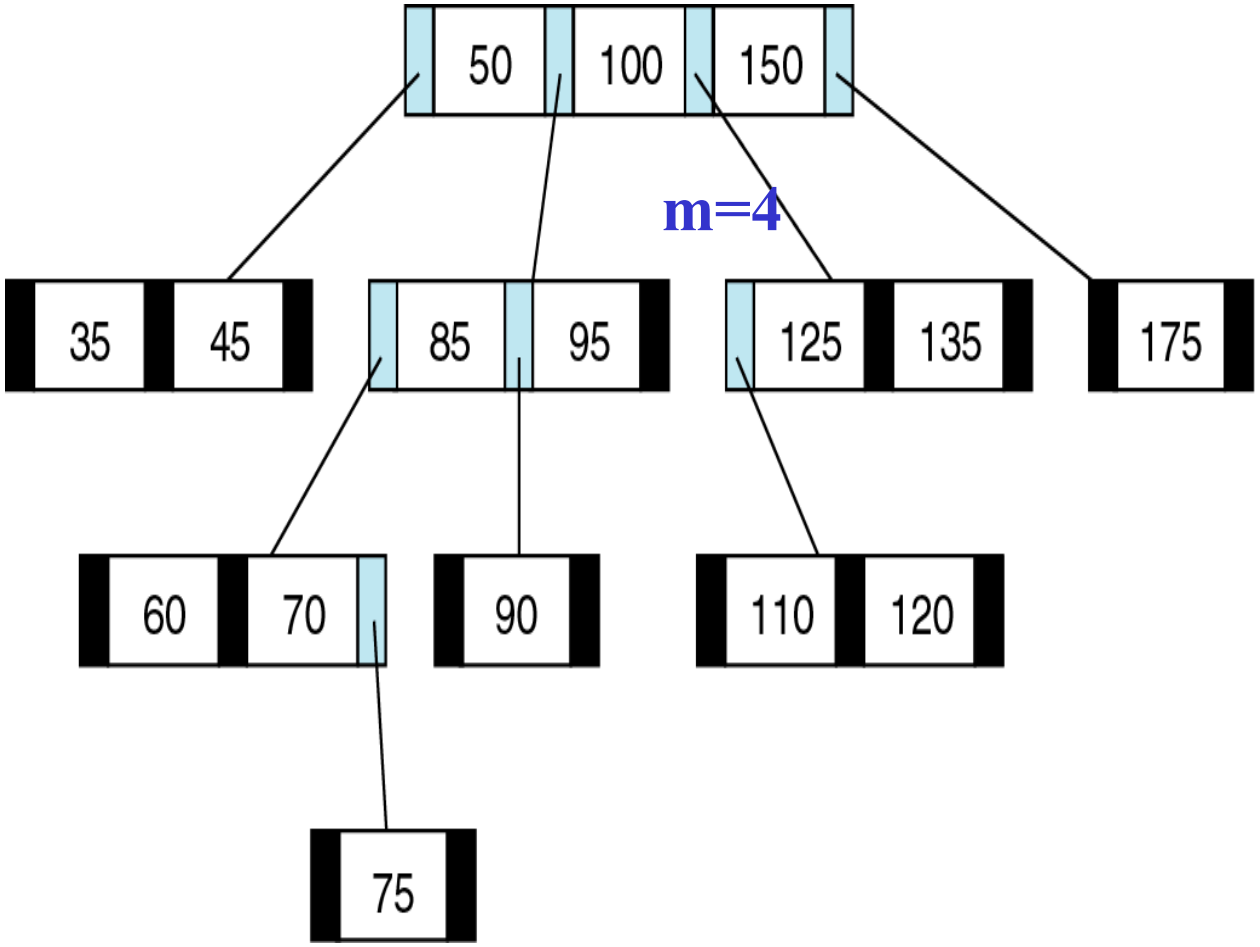
Given the nonempty multiway tree, the following properties are defined:

1. Each node has **0** to **m** subtrees.
2. Given a node contains **k** subtree pointers, some of which may be null, and **k-1** data entries.
3. The key values in the first subtree are all less than the key in the first entry, the key values in the other subtrees are all greater than or equal to the key in their parent entry.
4. The keys of the data entries are ordered.

Advance Tree

M-Way Trees





M-way tree of order 4. Key = (m-1)=3

Advance Tree (B-Tree)

Motivation for B-Trees

- Index structures for large datasets cannot be stored in main memory
- Storing it on disk requires different approach to efficiency
- Assuming that a disk spins at 3600 RPM, one revolution occurs in 1/60 of a second, or 16.7ms
- Crudely speaking, one disk access takes about the same time as 200,000 instructions.
- Assume that we use an AVL tree to store about 20 million records
- We end up with a **very** deep binary tree with lots of different disk accesses; $\log_2 20{,}000{,}000$ is about 24, so this takes about 0.2 seconds
- We know we can't improve on the **log*n*** lower bound on search for a binary tree
- But, the solution is to use more branches and thus reduce the height of the tree!
  - As branching increases, depth decreases

Advance Tree (B-Tree)

Introduction of B-Trees

- A B-tree index is a self-balancing tree data structure that keeps data sorted and allows searches, sequential access, insertions, and deletions in **logarithmic time.**
- The B-tree index structure is widely used in databases and file systems.
- B-Tree indexing is widely used in relational databases such as MySQL and PostgreSQL.
- A B-tree of order *m* is an ***m*-way** tree (i.e., a tree where each node may have up to *m* children) in which:
  1. Every node can have at most **m−1** keys and **m** children.
  2. All leaves are on the same level
  3. All nodes except the root have at least $\lceil m/2 \rceil$ children
  4. All nodes except the root have at least $\lceil m/2 \rceil - 1$ keys
  5. The root is either a leaf node, or it has from **two** to ***m*** children [min 1 key]
  6. Keys within a node are sorted.
- Time Complexity of B-Tree: **O(logn)** for search, insert and delete

Advance Tree (B-Tree)

Inserting into a B-Tree

- A new item must be added only at the leaf node only.

1. **IF** tree is **Empty**,
   1. Create a **newNode** with a given ***key*** value
   2. Insert **newNode** as a root node
2. **ELSE**
   1. LOOKUP the tree to find the suitable leaf node to which the ***key*** value can be added
   2. IF the leaf node has empty position
      1. Insert the ***key*** value to the right position in the leaf node (ascending order)
   3. IF the leaf node is already full
      1. **split** the leaf node by sending middle value to its parent node. Insert the ***key*** into the appropriate position in one of the new nodes you just got by splitting.
      2. Repeat the same if the parent is also full. This might have to be repeated all the way to the ***root***.
      3. If the splitting is performed at ***root*** node, then the middle value becomes new ***root*** node for the tree and the height of the tree is increased by one.

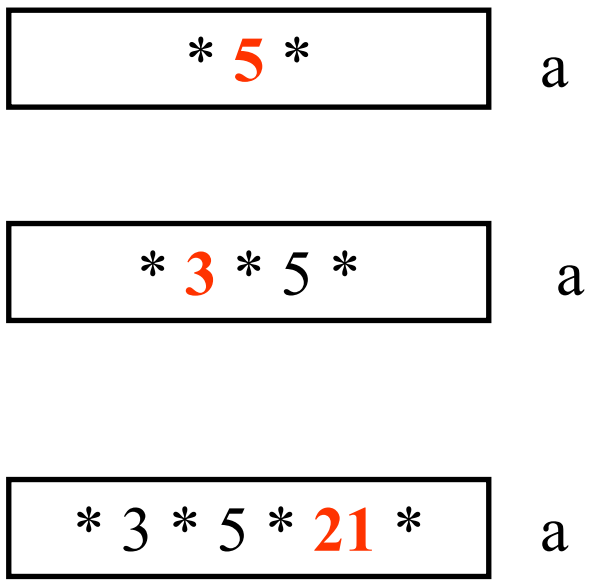
Advance Tree (B-Tree)

Insertion Operations

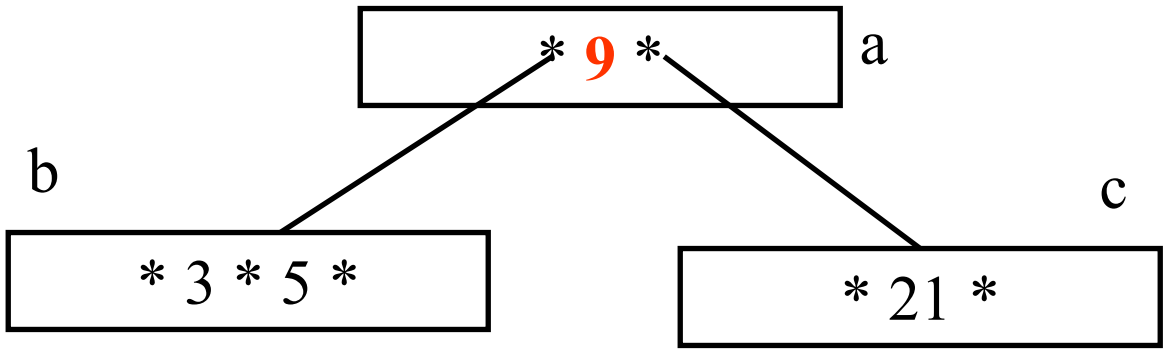
- B-Tree of order 4
  - Each node has at most 4 pointers and 3 keys, and at least 2 pointers and 1 key.
- Insert: 5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8
- Delete: 2, 21, 10, 3, 4

Advance Tree (B-Tree)

Insert 5, 3, 21



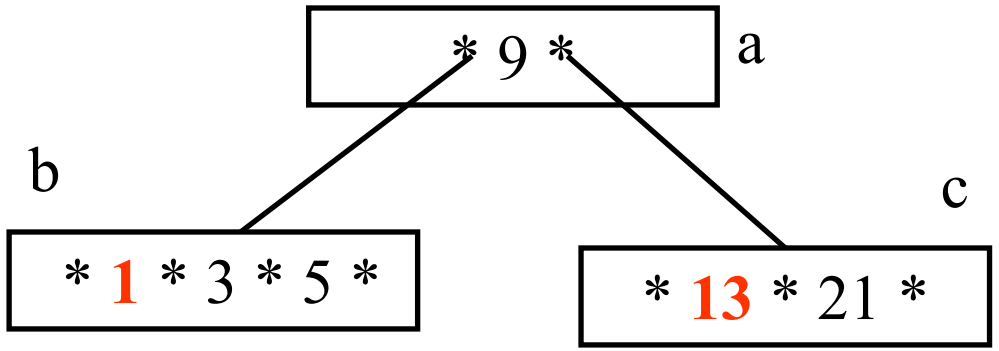
Insert 9



Node a splits creating 2 children: b and c

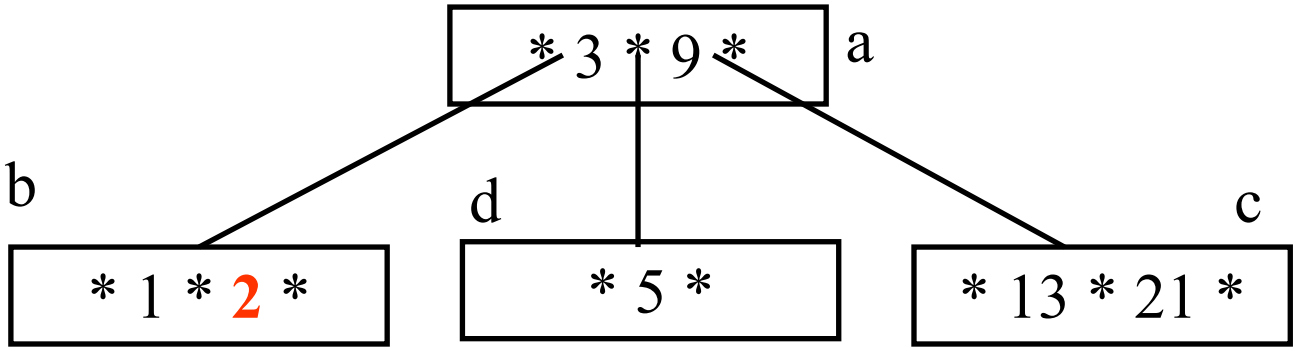
Advance Tree (B-Tree)

Insert 1, 13



Nodes b and c have room to insert more elements

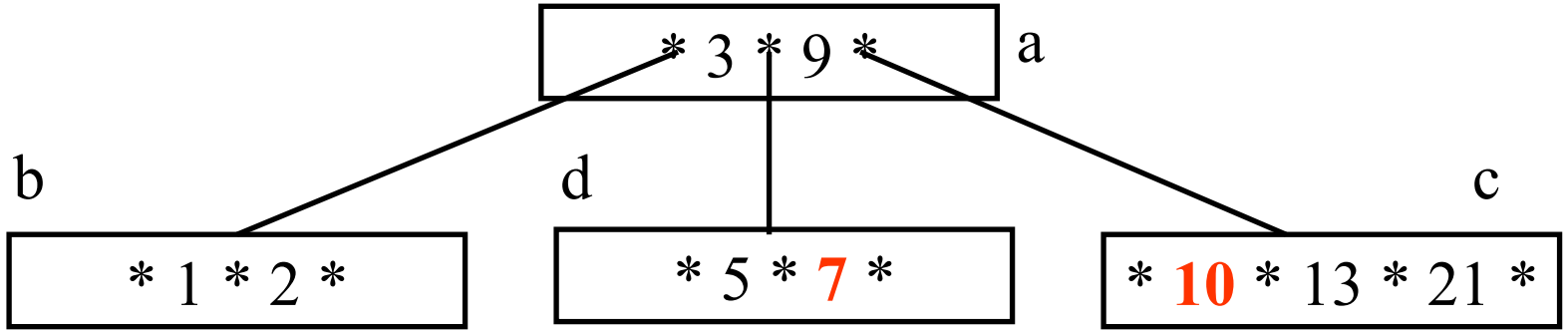
Insert 2



Node b has no more room, so it splits creating node d.

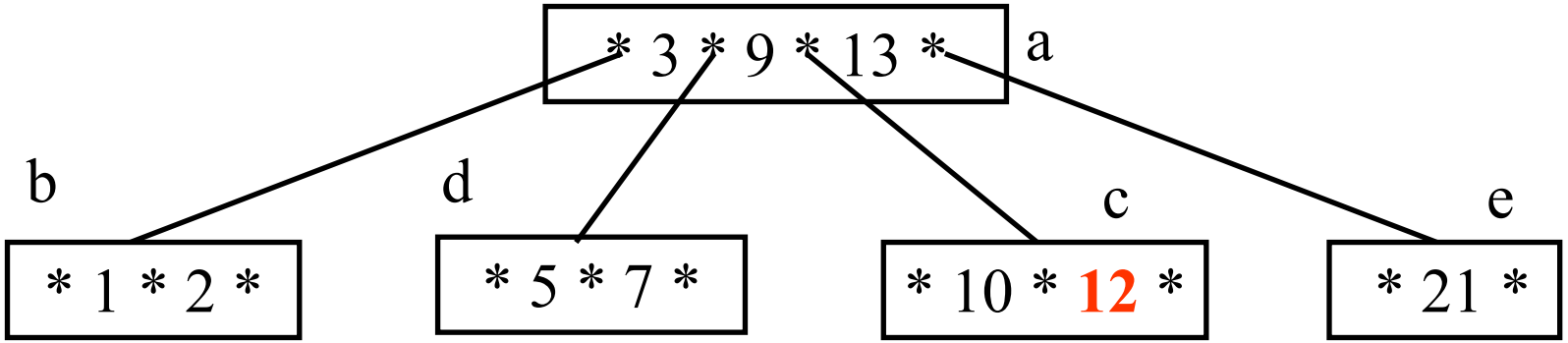
Advance Tree (B-Tree)

Insert 7, 10



Nodes d and c have room to add more elements

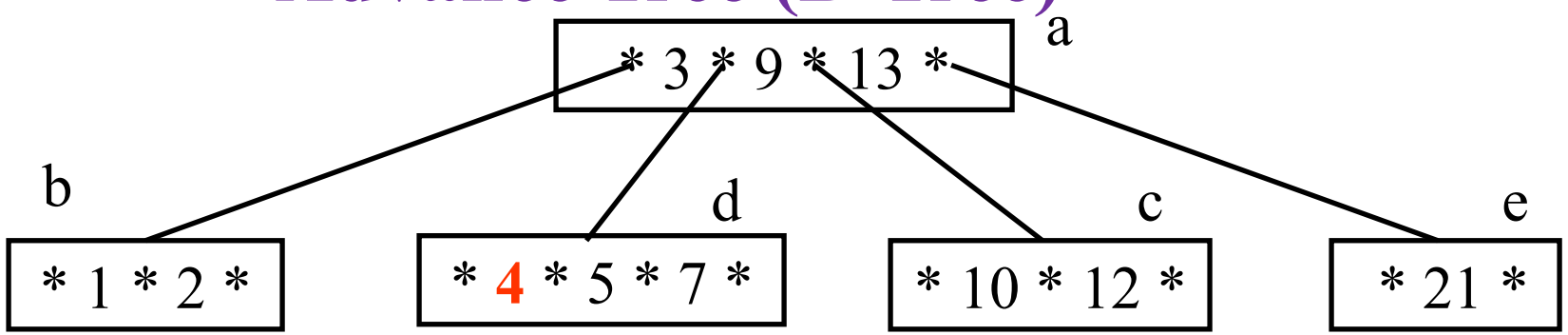
Insert 12



Nodes c must split into nodes c and e

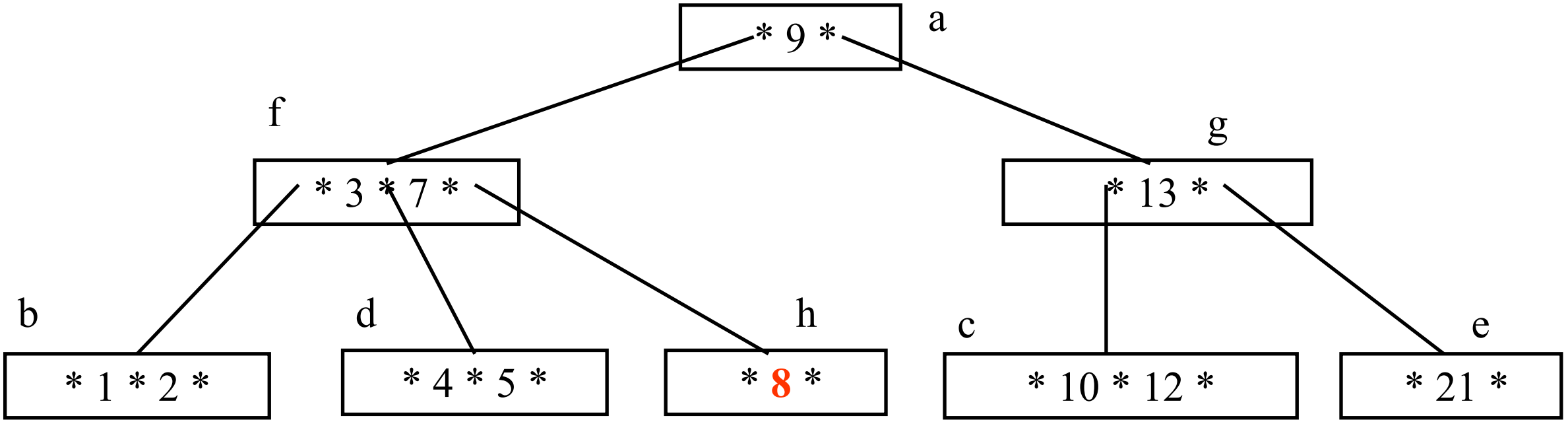
Advance Tree (B-Tree)

Insert 4



Node d has room for another element

Insert 8



Node d must split into 2 nodes. This causes node a to split into 2 nodes and the tree grows a level.

Advance Tree (B-Tree)

Try yourself

Construct a B-tree of order 5 from the following keys arrive in order :

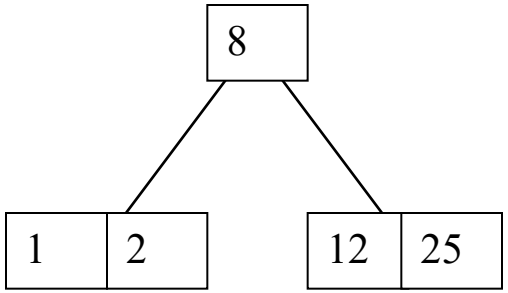
1 12 8 2 25 6 14 28 17 7 52 16 48 68 3 26 29 53 55 45

Advance Tree (B-Tree)

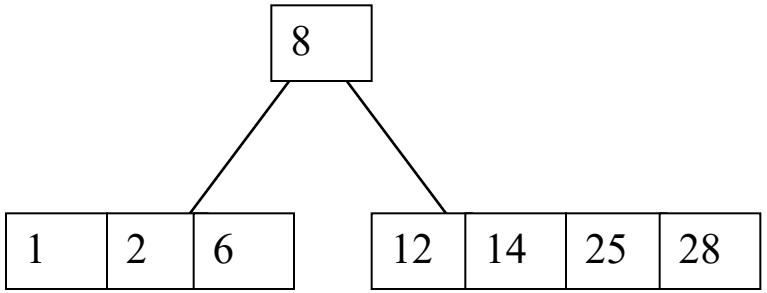
The first four items go into the root:

1	2	8	12

To put the fifth item in the root would violate condition 5
Therefore, when 25 arrives, pick the middle key to make a new root

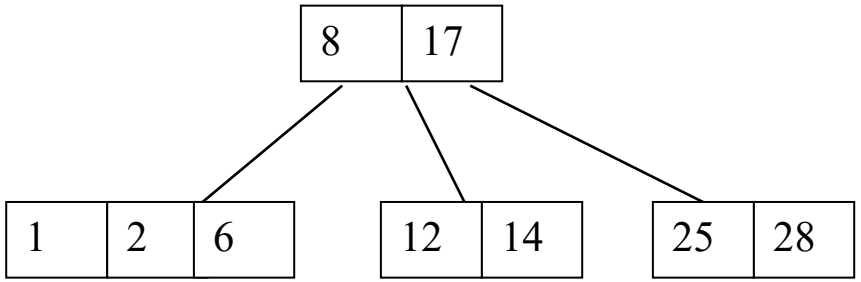


6, 14, 28 get added to the leaf nodes:

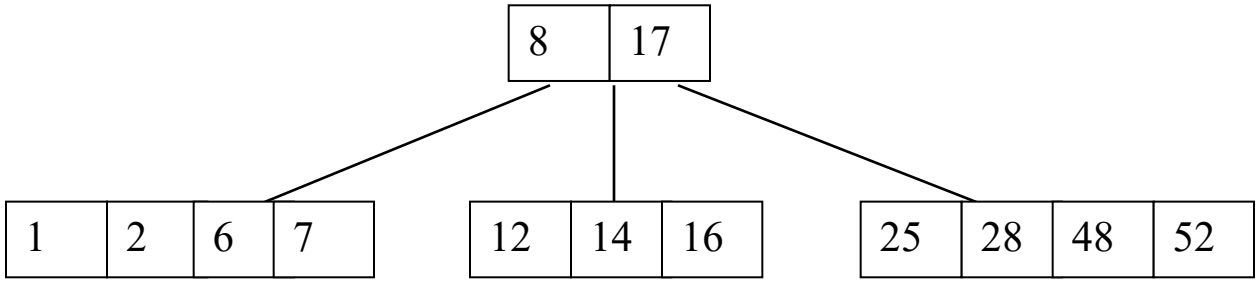


Advance Tree (B-Tree)

Adding 17 to the right leaf node would over-fill it, so we take the middle key, promote it (to the root) and split the leaf

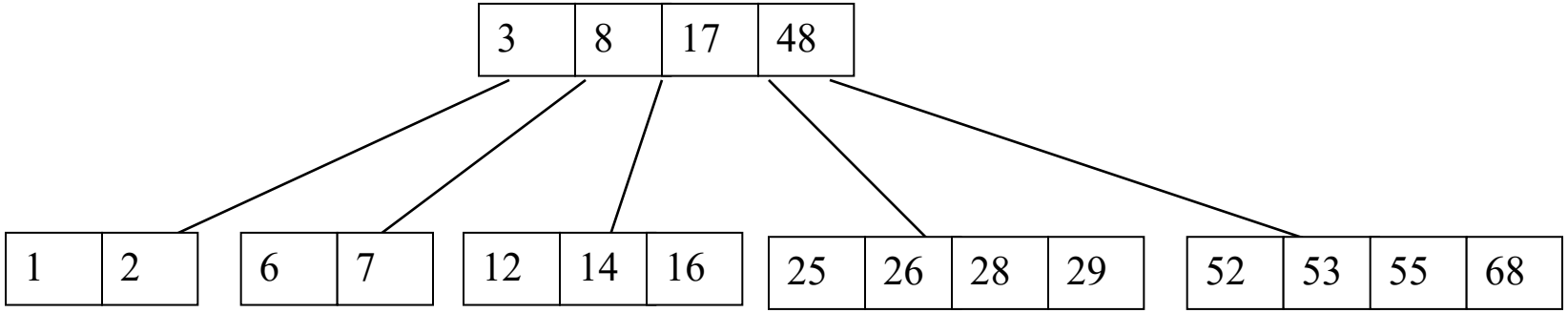


7, 52, 16, 48 get added to the leaf nodes

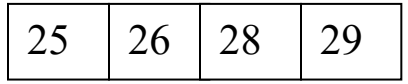


Advance Tree (B-Tree)

Adding 68 causes us to split the right most leaf, promoting 48 to the root, and adding 3 causes us to split the left most leaf, promoting 3 to the root; 26, 29, 53, 55 then go into the leaves



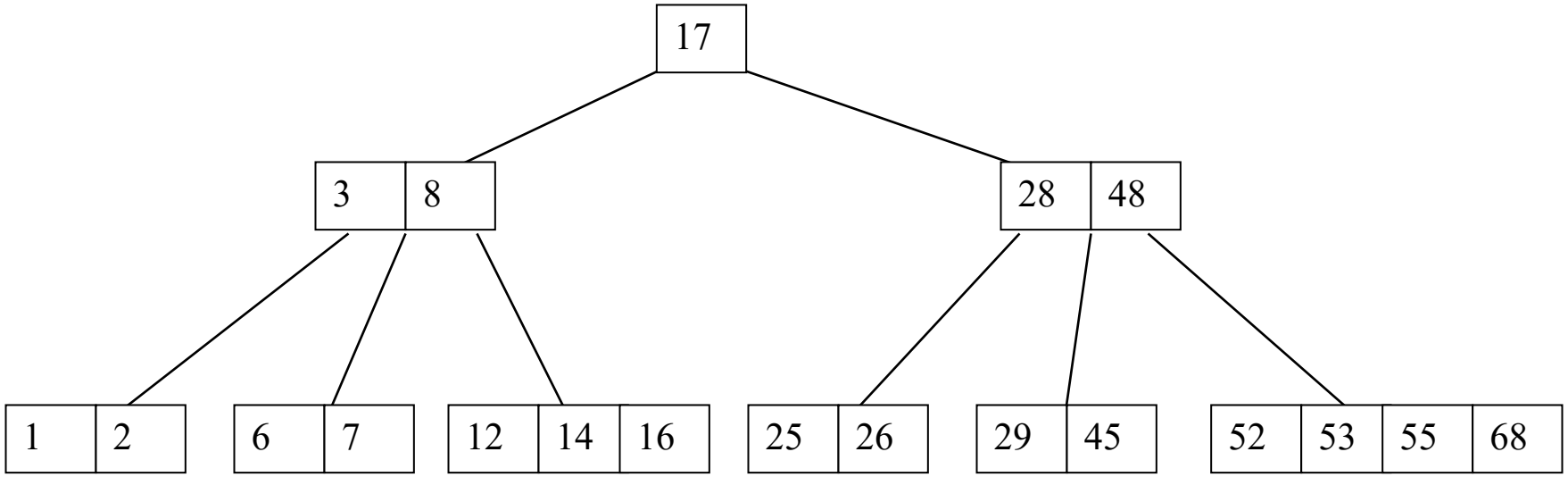
Adding 45 causes a split of



and promoting 28 to the root then causes the root to split

Advance Tree (B-Tree)

Final B-Tree of given try yourself



Advance Tree (B-Tree)

Removal from a B-tree (Delete Operation)

- During insertion, the ***key*** always goes into a leaf. For deletion we wish to remove from a leaf. There are three possible ways we can do this:
  1. If the ***key*** is already in a leaf node, and removing it doesn't cause that leaf node to have too few ***keys***, then simply remove the ***key*** to be deleted.
  2. If the ***key*** is not in a leaf then it is guaranteed (by the nature of a B-tree) that its predecessor or successor will be in a leaf - in this case we can delete the ***key*** and promote the predecessor or successor ***key*** to the non-leaf deleted ***key's*** position.
- If (1) or (2) lead to a leaf node containing less than the minimum number of ***keys*** then we have to look at the siblings immediately adjacent to the leaf in question:
  3. If one of them has more than the min. number of ***keys*** then we can promote one of its ***keys*** to the parent and take the parent ***key*** into our lacking leaf
  4. If neither of them has more than the min. number of ***keys*** then the lacking leaf and one of its neighbours can be combined with their shared parent (the opposite of promoting a ***key***) and the new leaf will have the correct number of ***keys***; if this step leave the parent with too ***few*** keys then we repeat the process up to the root itself, if required

B-Tree: Delete Operation

Input: A B-Tree *T* and a *key* to delete
Output: The B-Tree *T* with *key* removed

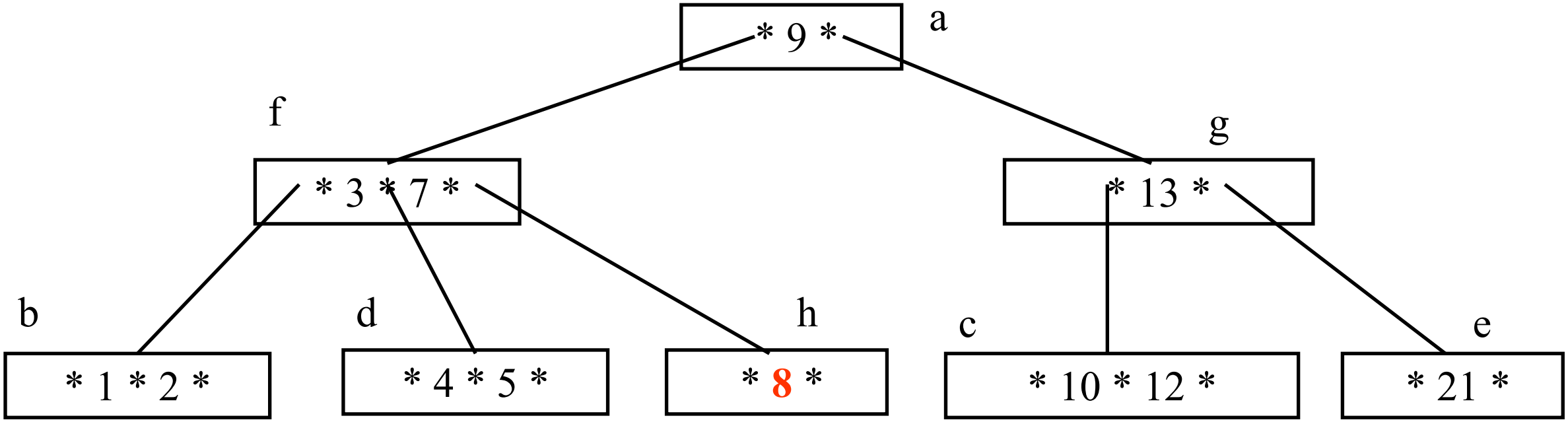
Algorithm *Deletion(T,key)*

1. IF *(search (T,key) == null)*, exit().
2. IF ***key*** is in leaf node *N*
   i. Remove the ***key*** from *N*
   ii. IF *(N.size() <⌈**m/2**⌉-**1**)*, underflow is occurred. **GOTO Step 4**.
3. IF ***key*** is in intermediate node *N*
   i. *replace(key,predecessor(key))* *//replace (key ,successor(key))*
   ii. *Deletion(T,key)* // recursive call to delete ***key*** from leave node
4. IF a left or right siblings has an extra ***key*** //Borrow the nearest *key* from sibling node
   i. Move a ***key*** from the parent down to the underflowing node
   ii. Move a ***key*** from the sibling up to the parent.
5. IF a left or right sibling has **NO** extra ***key*** //Borrow is not possible
   i. Merge the ***key*** to the left *sibling* (or merge with right *sibling*)
   ii. PULL a ***key*** from the parent down to the left *sibling* to merge
   iii. Check the parent for underflow.
   iv. IF parent has underflow, **GOTO Step 4** //handle it recursively if underflow occurred again for parent

Note:
Min keys per node(except root): **⌈m/2⌉-1**
Max keys per node: **m-1**
All leaves at same level

Advance Tree (B-Tree)

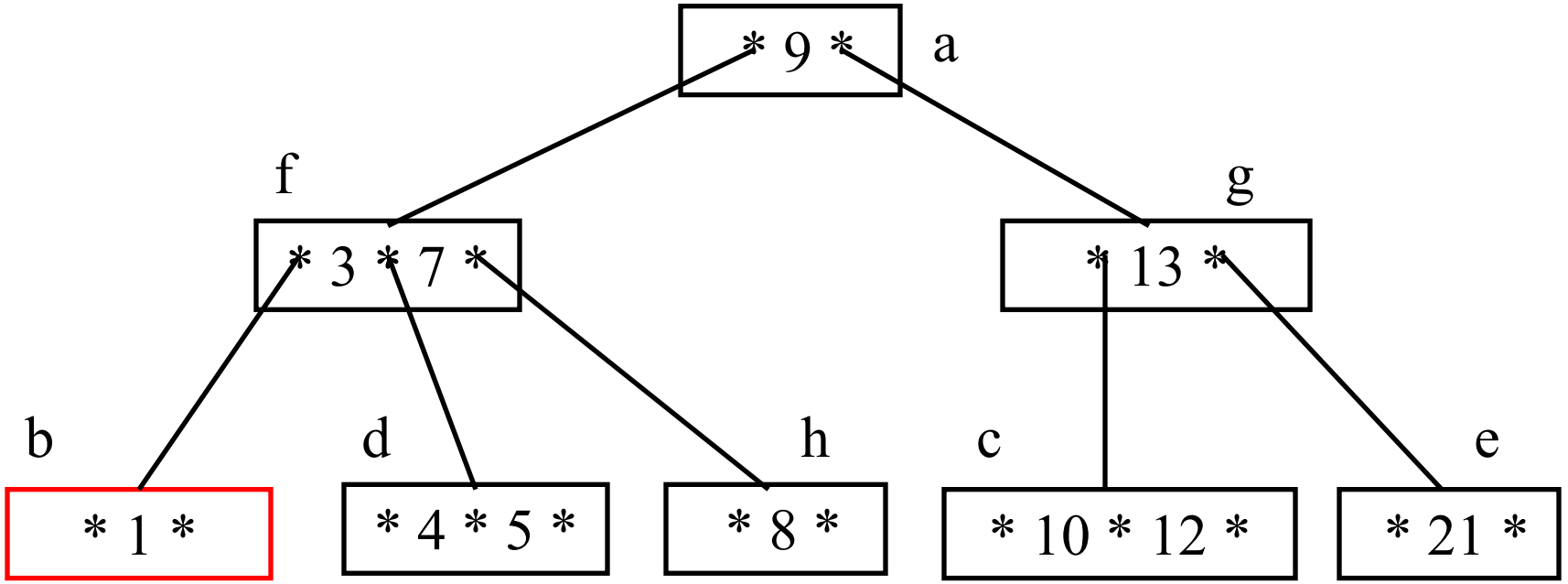
Delete: 2, 21, 10, 3, 4



B-Tree Deletion

Delete 2 (Step 2 of Algorithm)

Next: Delete 21

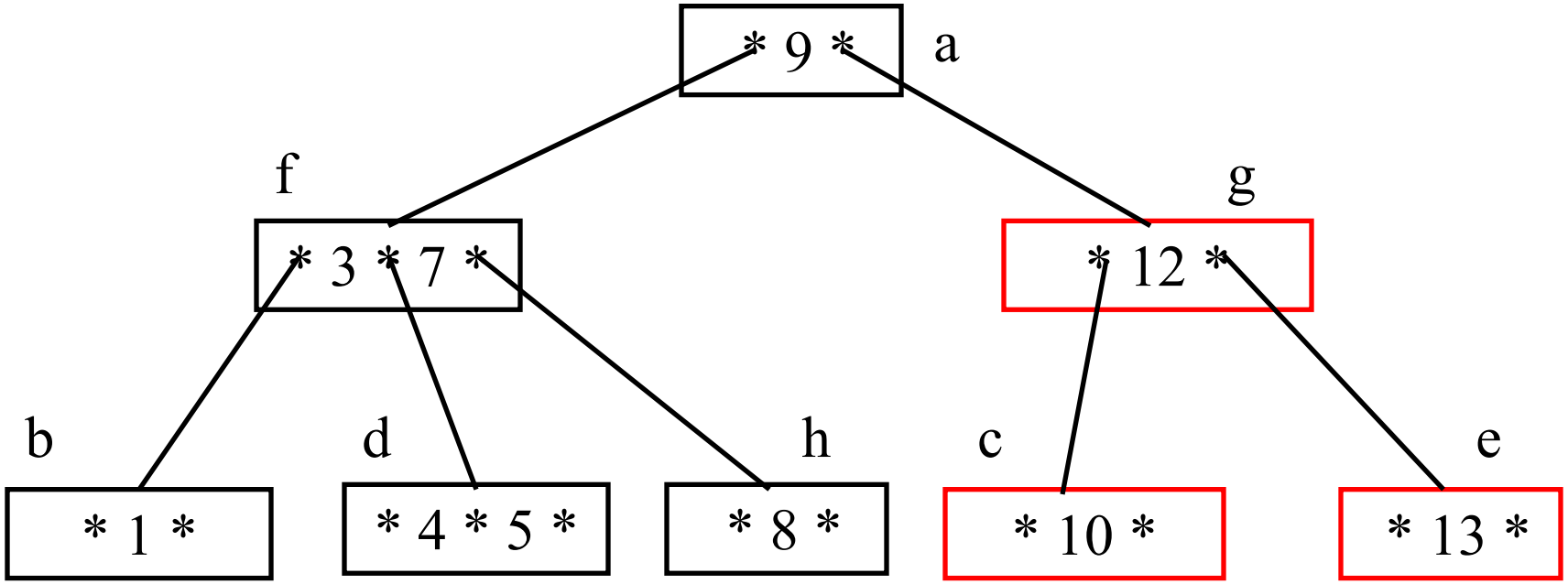


Node b can loose an element without underflow.

B-Tree Deletion

Delete 21 (Step 4 of Algorithm)

Next: Delete 10

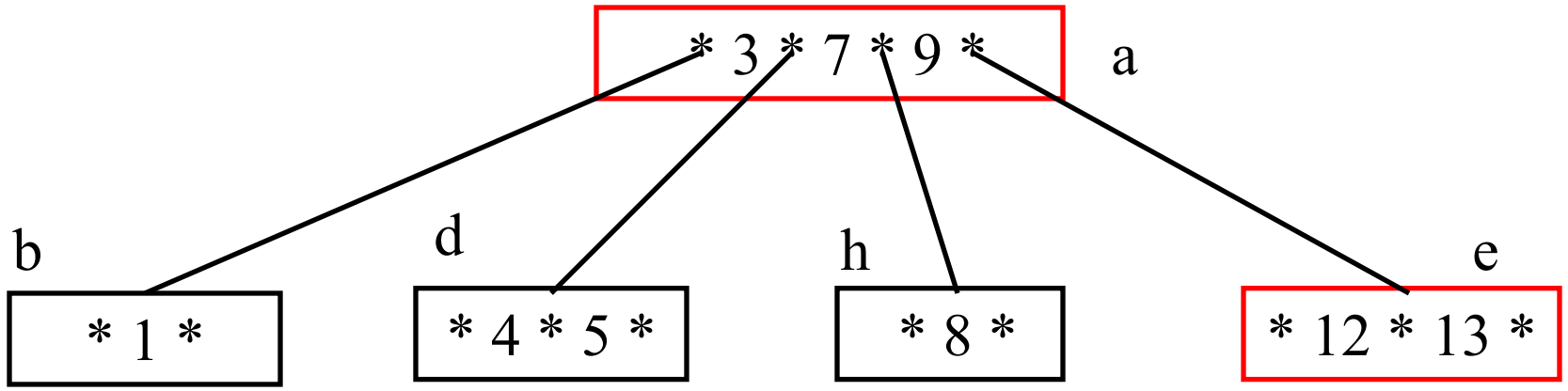


Deleting 21 causes node e to underflow, so elements are redistributed between nodes c, g, and e

B-Tree Deletion

Delete 10 (Step 5 of Algorithm)

Next: Delete 3

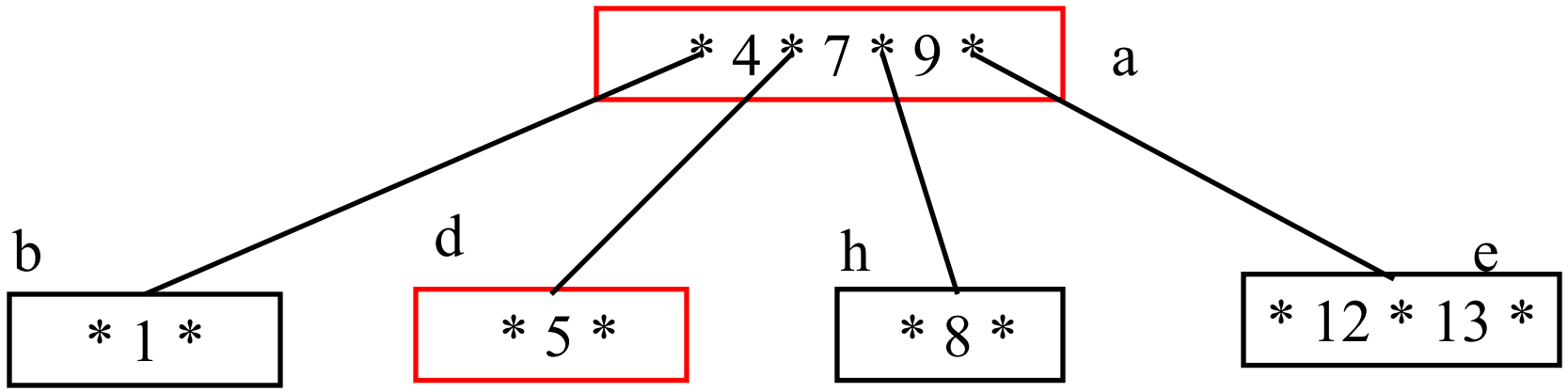


Deleting 10 causes node c to underflow. This causes the parent, node g to recombine with nodes f and a. This causes the tree to shrink one level.

B-Tree Deletion

Delete 3

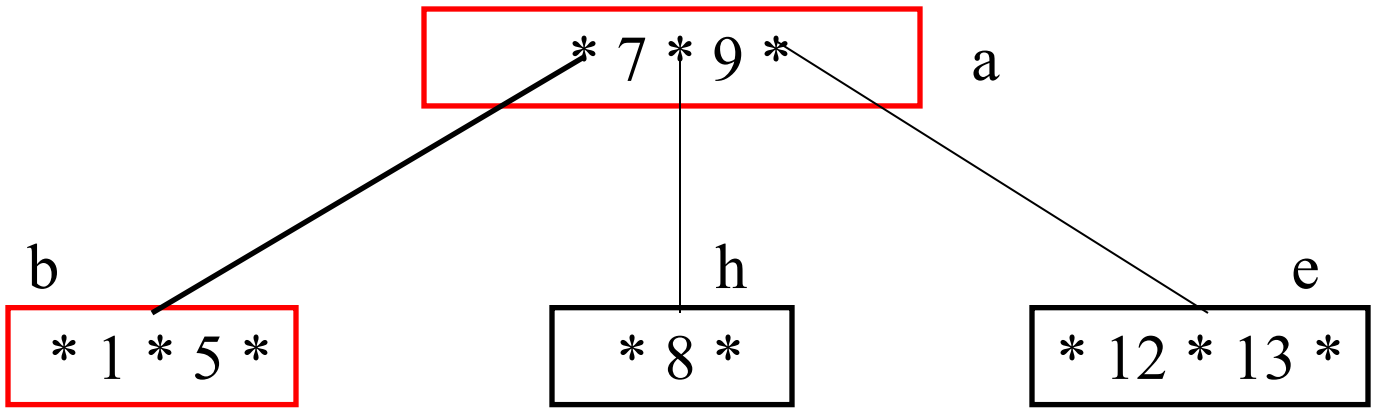
Next: Delete 4



Because 3 is a pointer to nodes below it, deleting 3 requires keys to be redistributed between nodes a and d.

B-Tree Deletion

Delete 4



Deleting 4 requires a redistribution of the keys in the subtrees of 4; however, nodes b and d do not have enough keys to redistribute without causing an underflow. Thus, nodes b and d must be combined.

Why B-Tree ?

- The need for **B-tree** arose with the rise in the need for lesser time in accessing the secondary storage (HDD).
- The secondary storage devices are slower with a larger capacity. There was a need for such types of data structures that *minimize the disk accesses*.
- Other data structures such as a *binary search tree*, AVL tree, red-black tree, etc can store only one key in one node. If we have to store a large number of keys, then the height of such trees becomes very large and the access time increases.
- B-tree can store many keys in a single node and can have multiple child nodes. *This decreases the height significantly allowing faster disk accesses*.
- Generally, the *B-Tree node size is kept equal to the disk block size*.

Applications of B-Tree

1. **Database Systems**
   - B-trees are commonly used to implement database indexing, allowing efficient access to large datasets. Used in systems like MySQL, PostgreSQL, and Oracle for indexing primary and secondary keys.
2. **File Systems**
   - Used to manage file systems where quick access to metadata (e.g., file locations) is critical.
   - Examples include NTFS (Windows) and HFS+ (macOS).
3. **Operating Systems**
   - For managing paging and virtual memory, where fast lookup of page tables is required.
   - Helps in organizing data structures used by kernels.
4. **Multilevel Indexing**
   - B-trees support multi-level indexing, making them ideal for hierarchical data storage, such as directory structures.
5. **Data Warehousing and Analytics**
   - Useful for indexing large datasets, especially for OLAP (Online Analytical Processing) systems.
6. **Search Engines**
   - To index and search massive amounts of data efficiently.
7. **Geographic Information Systems (GIS)**
   - For managing spatial data where multidimensional range queries are performed.
8. **Networking**
   - Applied in routing tables for fast lookup and update of IP routing entries.

B+ Tree

- B+ tree is an extension of the B tree.
- In B tree the keys and records can be stored as internal as well as **leaf nodes** whereas in B+ trees, the records are stored as **leaf nodes** and the **keys are stored only in internal nodes**.
- The leaf nodes are also linearly connected in B+ trees to improve range-query performance.
- B+ tree indices are an alternative to indexed-sequential files.
- Disadvantage of indexed-sequential files
  - performance degrades as file grows, since many overflow blocks get created.
  - Periodic reorganization of entire file is required.

Advantage of B+tree index files:

- automatically reorganizes itself with small, local, changes, in the face of insertions and deletions.
- Reorganization of entire file is not required to maintain performance.
- (Minor) disadvantage of B+ trees: extra insertion and deletion overhead, space overhead.

B+ Tree Index Files

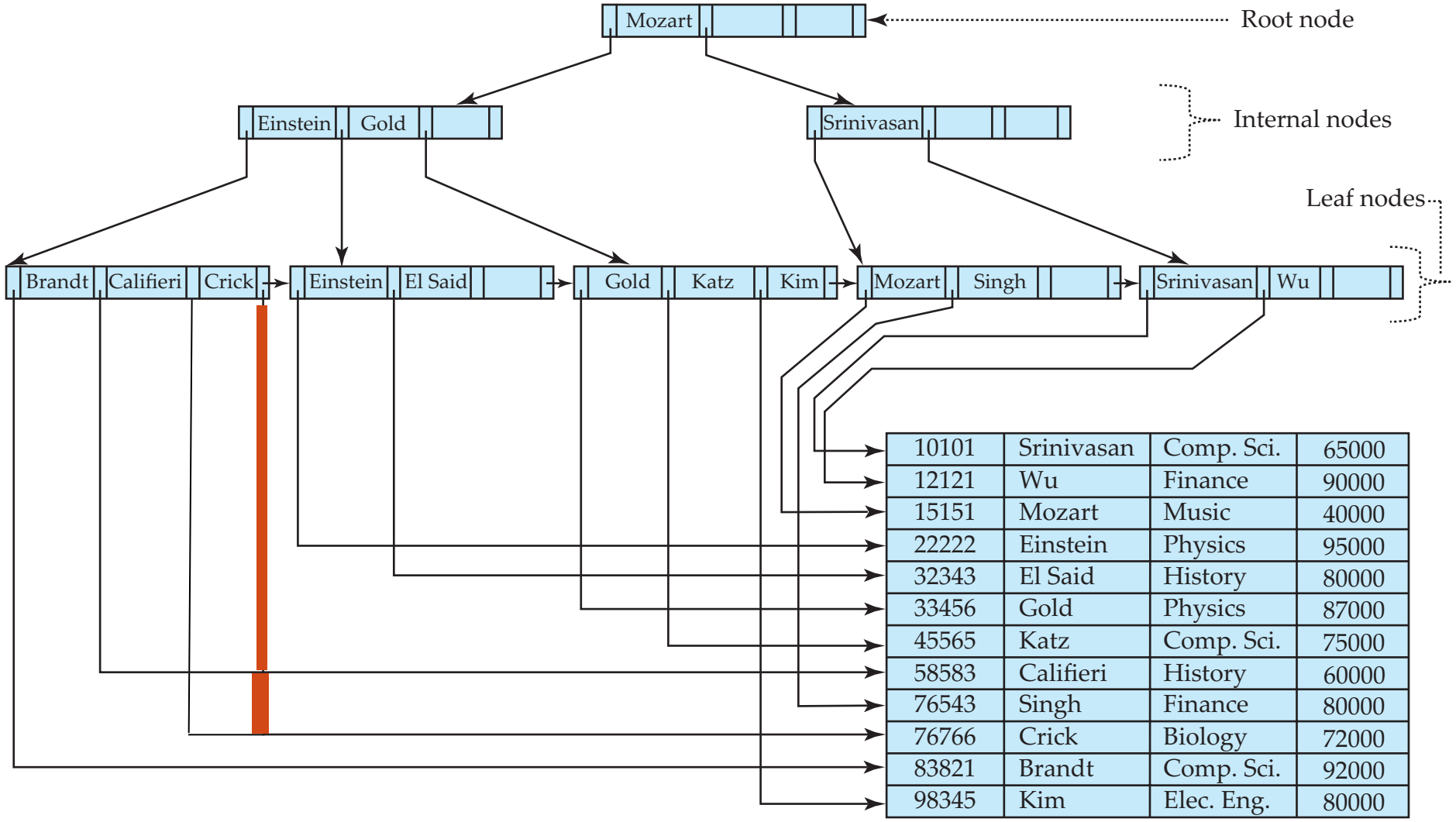
Typical node

P_1	K_1	P_2	. . .	P_{n-1}	K_{n-1}	P_n

- K_i are the search-key values
- P_i are pointers to children (for non-leaf nodes) or pointers to records or buckets of records (for leaf nodes).

- The search-keys in a node are ordered
  - $K_1 < K_2 < K_3 < . . . < K_{n-1}$
- Usually the size of a node is that of a block
- **A B+ tree is a rooted tree satisfying the following properties:**
  - *All paths from root to leaf are of the same length*
  - *Each node that is not a root or a leaf has between* $\lceil$***m/2***$\rceil$ *and* ***n*** *children.*
  - *A leaf node has between* $\lceil$***m/2***$\rceil$***-1*** *and* ***m–1*** *keys*
  - *Special cases:*
    - *If the root is not a leaf, it has at least* ***2*** *children.*
    - *If the root is a leaf (that is, there are no other nodes in the tree), it can have between* ***0*** *and* ***(m-1)*** *keys.*
  - *All values must be at leaf node.*

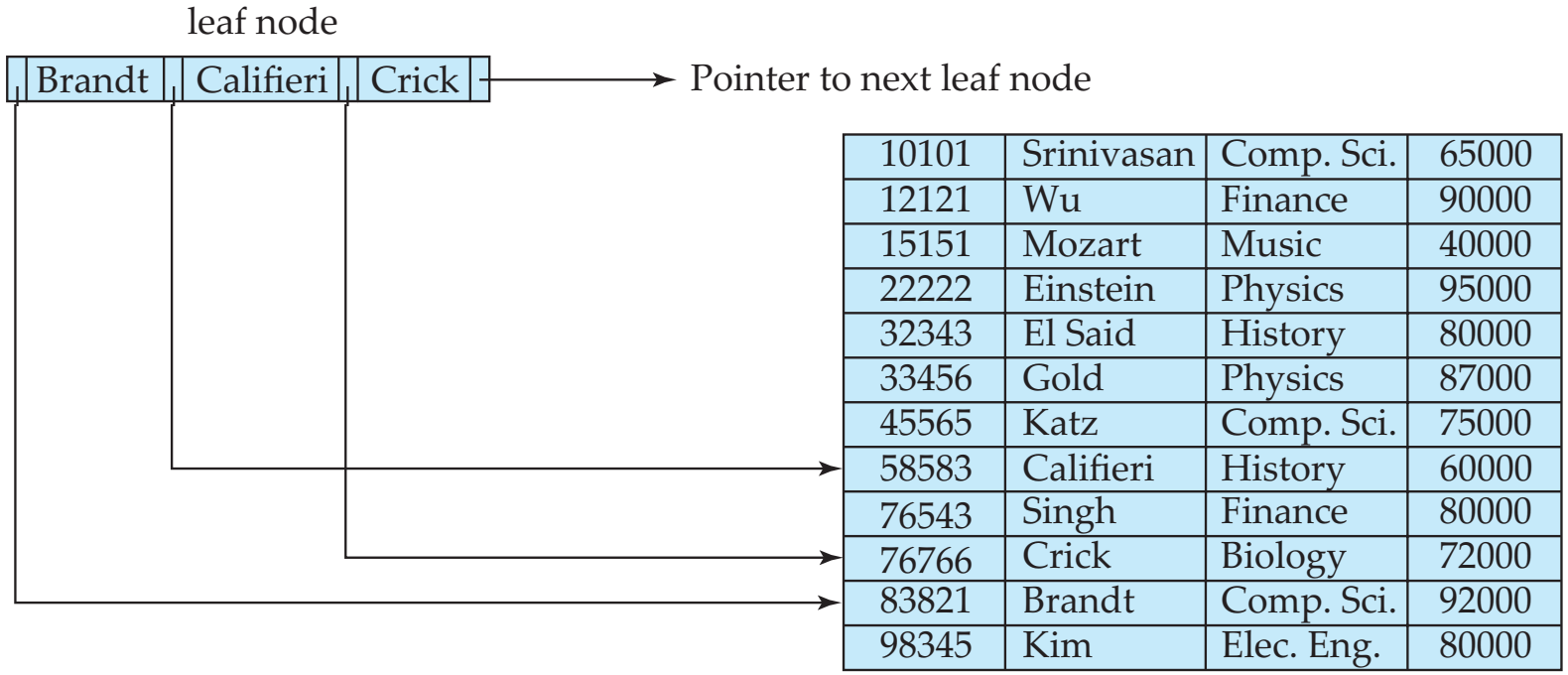
Example of B+-Tree



Leaf Nodes in B+-Trees

Properties of a leaf node:

- For $i = 1, 2, \ldots, n-1$, pointer P_i points to a file record with search-key value K_i,
- If L_i, L_j are leaf nodes and $i < j$, L_i's search-key values are less than L_j's search-key values
- P_n points to next leaf node in search-key order



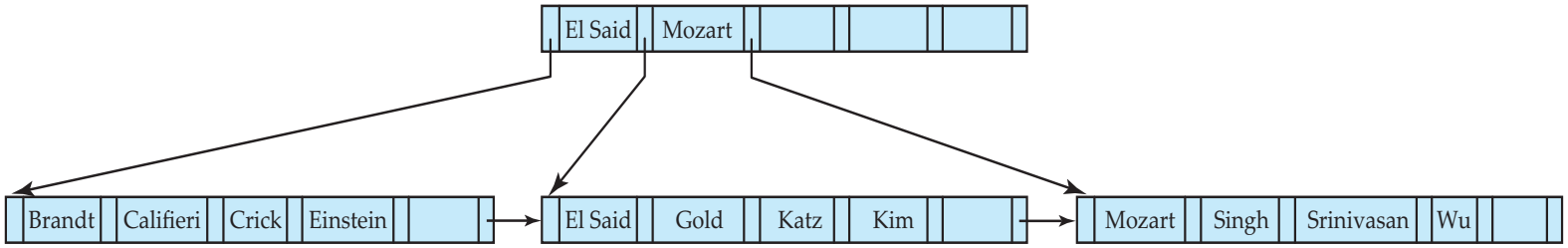
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	80000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	60000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Non-Leaf Nodes in B+-Trees

- Non leaf nodes form a multi-level sparse index on the leaf nodes. For a non-leaf node with m pointers:
  - All the search-keys in the subtree to which P_1 points are less than K_1
  - For $2 \leq i \leq m - 1$, all the search-keys in the subtree to which P_i points have values greater than or equal to K_{i-1} and less than K_i
  - All the search-keys in the subtree to which P_n points have values greater than or equal to K_{n-1}
  - General structure

P_1	K_1	P_2	…	P_{n-1}	K_{n-1}	P_n

- Example of B+-tree (m = 6)



- Leaf nodes must have between 3 and 5 values ($\lceil (m-1)/2 \rceil$ and $m - 1$, with m=6).
- Non-leaf nodes other than root must have between 3 and 6 children ($\lceil m/2 \rceil$ and m with m =6).[*for non leaf node: min no of key* = $\lceil m/2 \rceil$-1]
- Root must have at least 2 children.

Observations about B+-trees

- Since the inter-node connections are done by pointers, "logically" close blocks need not be "physically" close.
- The non-leaf levels of the B^+-tree form a hierarchy of sparse indices.
- The B^+-tree contains a relatively small number of levels
  - Level below root has at least $2 * \lceil m/2 \rceil$ values
  - Next level has at least $2 * \lceil m/2 \rceil * \lceil m/2 \rceil$ values
  - .. etc.
  - If there are K search-key values in the file, the tree height is no more than $\lceil \log_{\lceil m/2 \rceil}(K) \rceil$
  - thus searches can be conducted efficiently.
- Insertions and deletions to the main file can be handled efficiently, as the index can be restructured in logarithmic time.

B-tree vs. B+ tree (Order = m)[Min & Max Keys and Children]

Node Type	Tree Type	Min Keys	Max Keys	Min Children	Max Children
Internal (non-root)	B-tree	⌈m/2⌉ – 1	m – 1	⌈m/2⌉	m
	B+ tree	⌈m/2⌉ – 1	m – 1	⌈m/2⌉	m
Leaf	B-tree	⌈m/2⌉ – 1	m – 1	-	-
	B+ tree	⌈m/2⌉ - 1	m – 1	-	-
Root (special case)	Both	At least *1 key* if leaf, *2 children* if internal	m – 1	2 (if internal)	m

Updates on B+Trees: Insertion

1. Every element is inserted into a leaf node. So, go to the appropriate leaf node.
2. Insert the key into the leaf node in increasing order only if there is no overflow. If there is an overflow follow the steps to maintain the B+ Tree properties.

Properties for insertion B+ Tree

- **Case 1: Overflow in leaf node**
  1. Split the leaf node into two nodes.
  2. First node contains ⌈**(m-1)/2**⌉ values.
  3. Second node contains the remaining values.
  4. Copy the smallest search key value from second node to the parent node.(Right biased)
- **Case 2: Overflow in non-leaf node**
  1. Split the non leaf node into two nodes.
  2. First node contains **[m/2] - 1** keys and **[m/2]** children
  3. Second node contains **remaining** keys
  4. Move the smallest among remaining to the parent.

Updates on B+Trees: Insertion

Algorithm Insertion(T, key, value)

IF *(search(T, key) ≠ null), exit()* //Key already exists – no duplicates allowed (or update the value if needed)

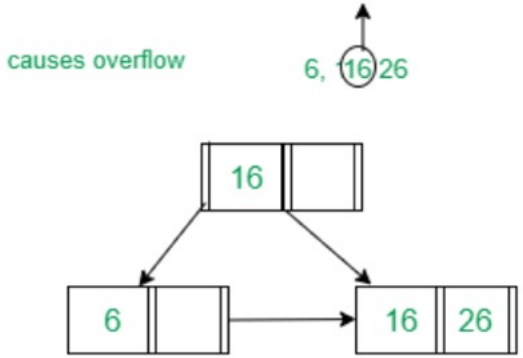
1. Find the appropriate leaf node L to insert the ***key***.
   1. Insert ***key*** and value into L in sorted order.
2. IF (L.size() ≤ m – 1), insertion is complete. Exit. *// No overflow– the leaf can hold the **key***
3. IF (L.size() > m – 1), overflow has occurred.
   Split L into two leaf nodes: Left node gets *⌊(m – 1)/2⌋* **keys** and Right node gets the remaining keys
   i. Copy the first *key* of the right node (k') up to the parent internal node (do not remove it from the leaf).
   ii. Insert k' into the parent internal node and add a pointer to the new right leaf.
   iii. Update linked list pointers between leaf nodes
4. IF (parent node ***size <= m – 1***) ,insertion is complete. Exit. *// No overflow*
5. IF (parent node ***size > m – 1***), overflow occurs again. Split for parent.// Split node into two internal nodes:
   Left internal node:
   keys = first ⌈m/2⌉ – 1
   children = first ⌈m/2⌉
   Right internal node:
   keys = remaining keys (after left)
   children = remaining child pointers (after left)
   Move the smallest key of the right node up to the parent
6. IF parent overflows, repeat Step 5 recursively up to root.

Updates on B+ Trees: Insertion (Example)

- Insert the following key values 6, 16, 26, 36, 46 on a B+ tree with order = 3.
- Here Key = m-1 = 3-1 =2 and Maximum no of childs = 3 (order is 3)

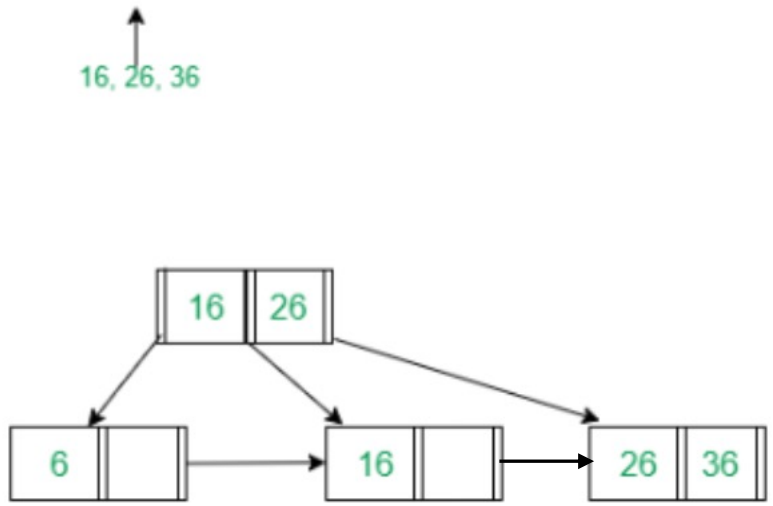
Insert 6, 16

Insert 26



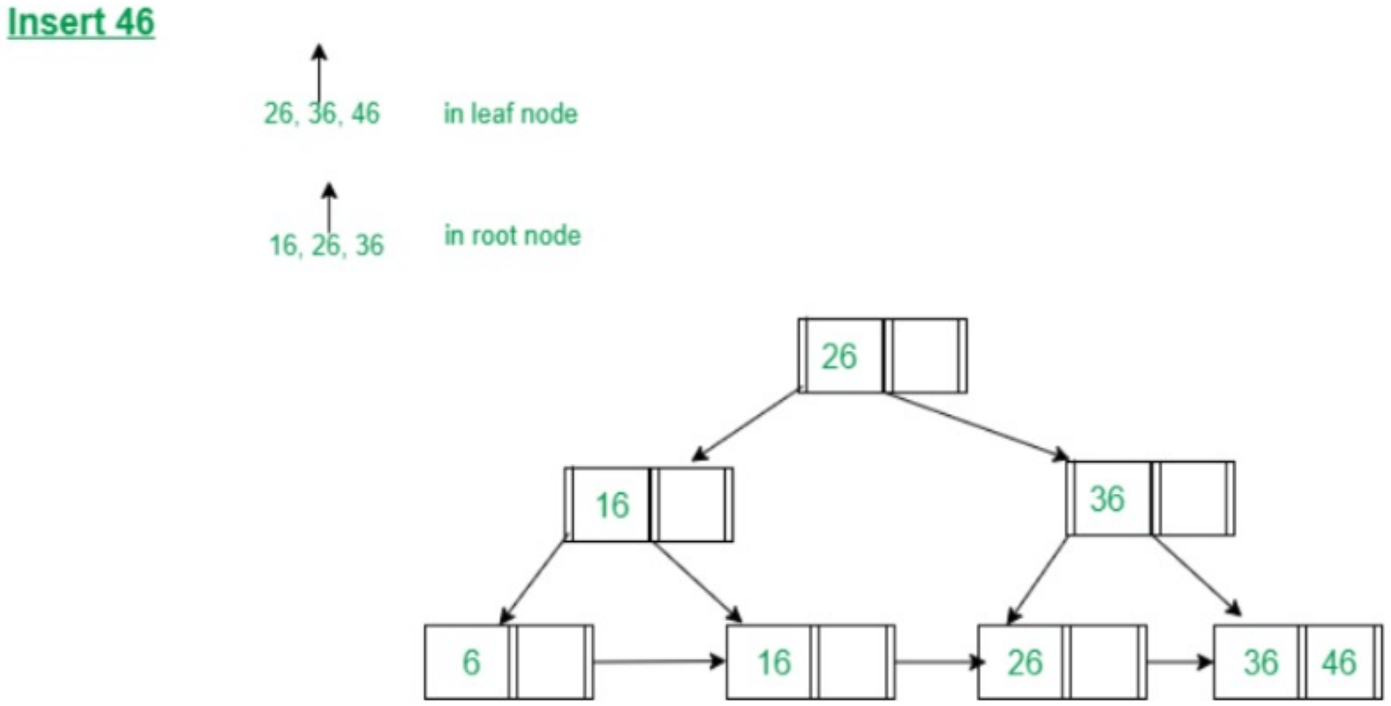
Inserting **26** cause overflow in the leaf node, so split it

Insert 36



Inserting **36** also cause overflow in the leaf node, so split it

Updates on B+-Trees: Insertion

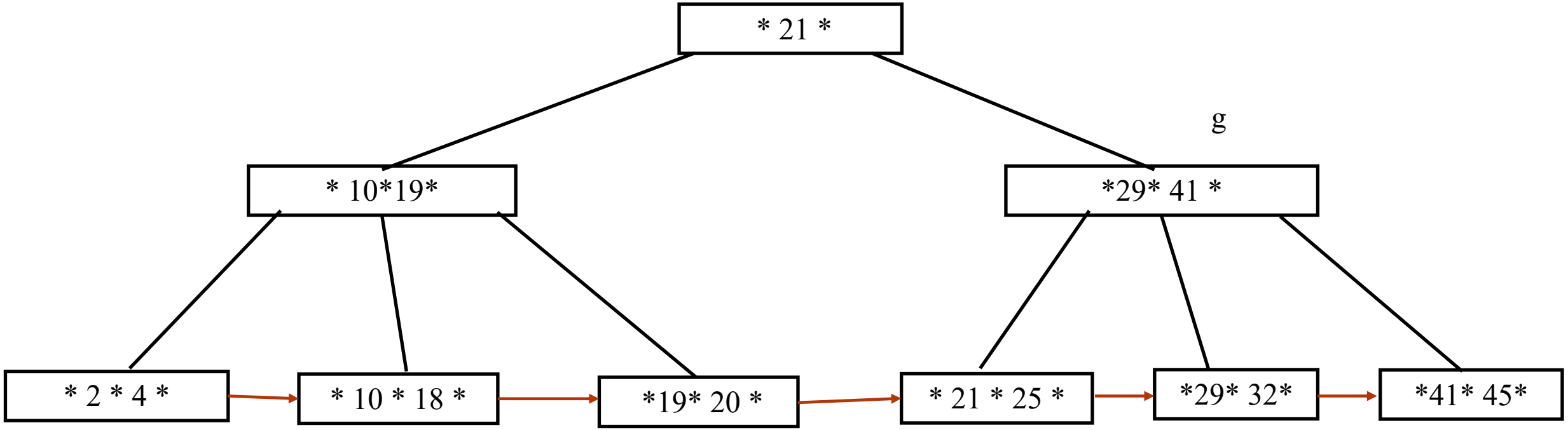


Inserting **46 again cause an overflow** in the leaf node. So, Split it and move the second part into its parents node which again cause an overflow in non leaf node. Apply the rule of splitting in non leaf node and maintain B+ tree property.

Try yourself

Create a B+ tree of order 4 for following numbers. 2, 4, 10, 18, 21, 32, 25, 19, 20, 29, 41,45

Step 1: Sort the data: 2,4,10,18,19,20,21,25,29,32,41,45



Updates on B+-Trees: Deletion

Delete a key from the leaf node and maintain B+ tree properties:

Case 1: Key not found

Case 2: Key is in a leaf node, and deletion does not cause underflow

Case 3: Key is in a leaf node, and deletion causes underflow, borrow possible

Case 4: Key is in a leaf node, and borrowing is not possible

Case 5: Key is in a leaf and internal node

Case 6: Root underflow

Algorithm Deletion(T, key) from B+ Tree

Algorithm Deletion(T, key)

IF *(search(T, key) == null)*, exit()

// Key does not exist – nothing to delete

// Step 1: Determine where the key exists

IF key is found in both an internal node I and leaf node L

GOTO Case 2b

ELSE IF key is found only in leaf node L

GOTO Case 2a

Case 2a: Key is in the leaf node only

1. Remove the key (and associated value) from leaf node L in sorted order
2. IF (L.size() ≥ ⌈m/2⌉-1

   // Leaf still has enough keys – no underflow

   Exit

   ELSE

   //Underflow occurred – go to Step 4

Case 2b: Key is in both internal node I (as separator) and leaf node L

1. Remove the key from leaf node L
2. IF (L.size() ≥ ⌈m/2⌉-1

   a. Check if the separator key in internal node I is still valid

   IF valid (i.e., still separates subtrees correctly)

   // Do nothing

   ELSE

   Replace the separator key in I with the new smallest key from right child

   b. Exit

ELSE

// Underflow occurred – go to Step 4

Algorithm Deletion(T, key) from B+ Tree

4. Handle Underflow in Leaf Node

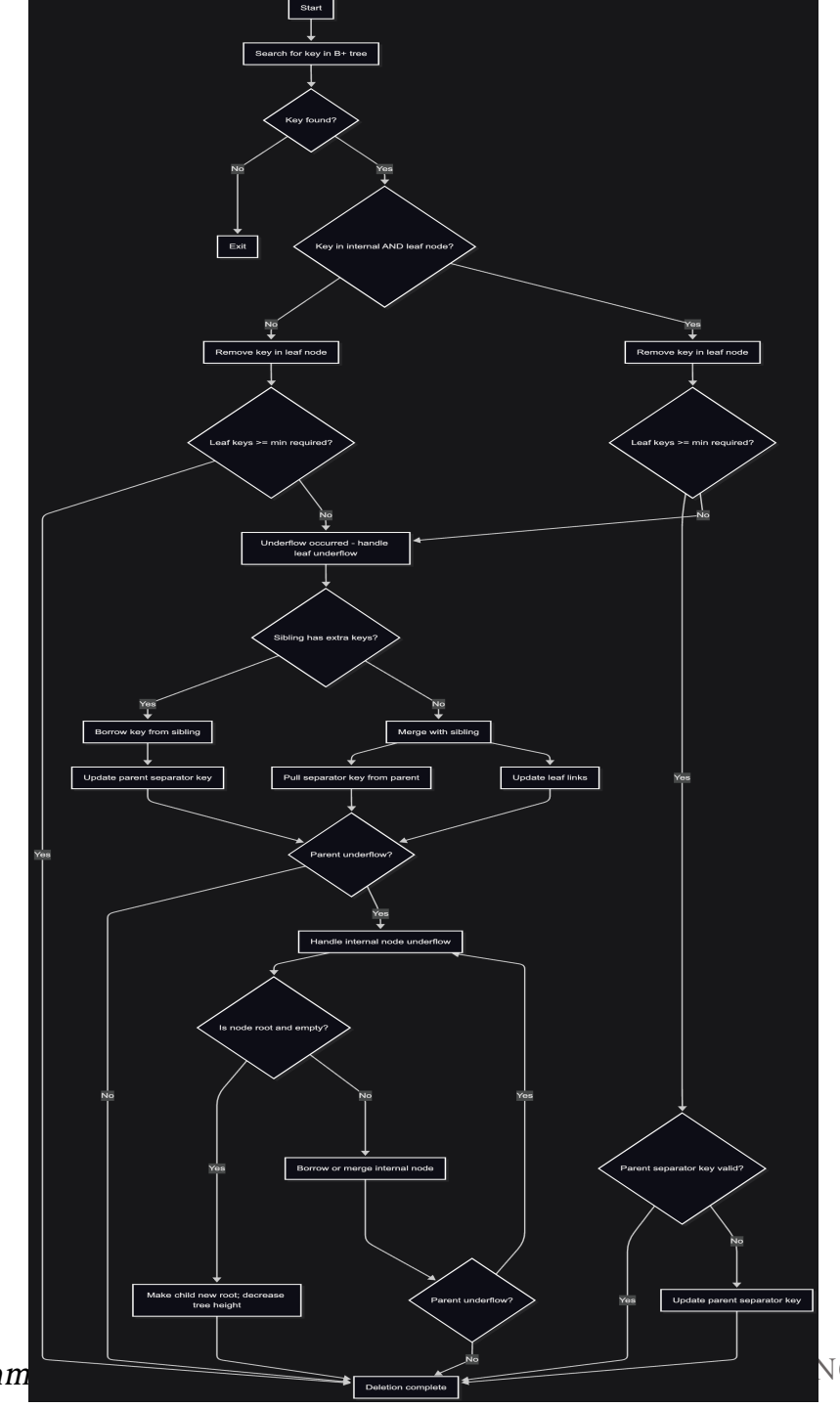
i. IF left or right sibling has more than ⌈m/2⌉-1 keys // Borrow from sibling
   i. Move one ***key*** from the sibling to L
   ii. Update parent ***key*** (since separator keys change)
ii. ELSE // No sibling has extra key – Borrowing not possible // Merge with sibling
   i. Merge L with its left or right sibling
   ii. Pull the separator ***key*** from parent down to complete the merge
   iii. Update linked list pointer between leaf nodes
   iv. Check parent for underflow. If underflow, GOTO Step 5

5.Handle Underflow in Parent Internal Node

i. IF (parent.size() ≥ ⌈m/2⌉ – 1), done
ii. ELSE
   i. Borrow from sibling or merge with sibling
   ii. Pull ***key*** from grandparent and continue recursively if needed
   iii. IF root node becomes empty (has no keys and only one child)
   iv. Make the single child the new root
   v. Tree height decreases by 1

6. Special Case: Root Underflow

IF root node becomes empty (has no keys and only one child)
   Make the single child the new root
   Tree height decreases by 1

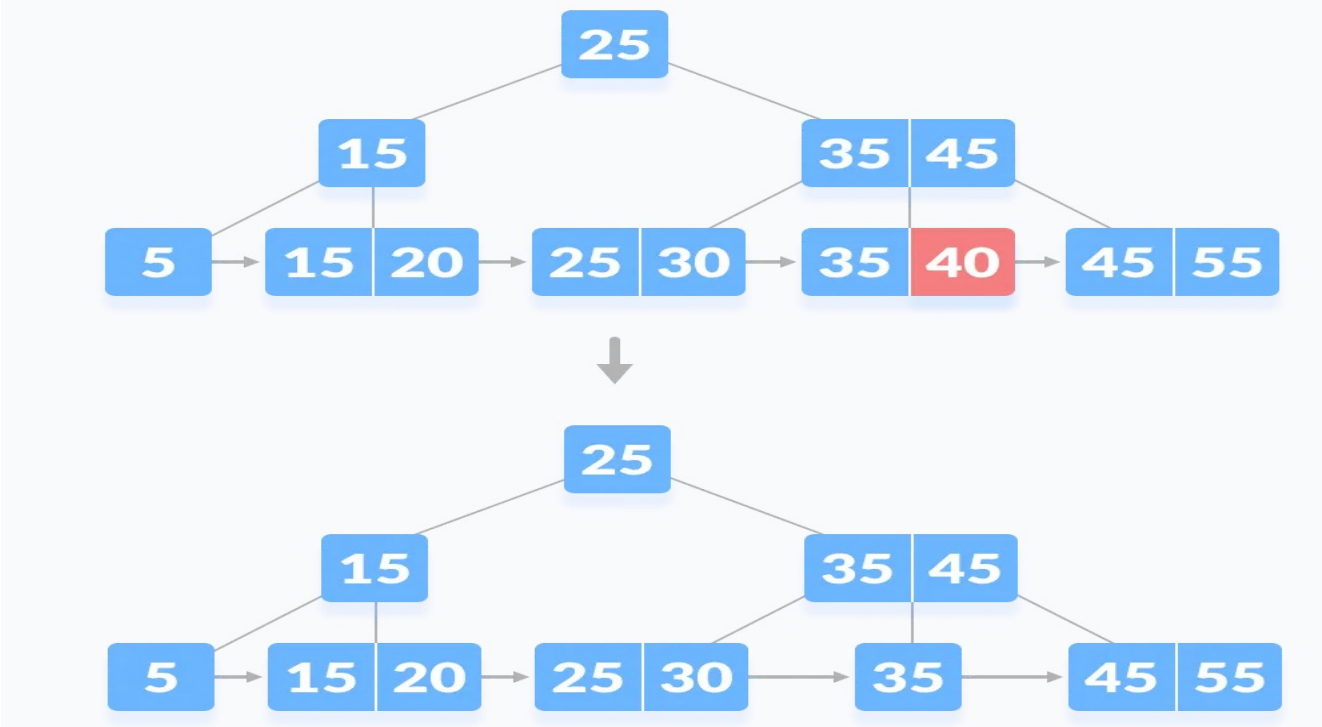


Flowchart of delete from B+ tree
Generated by: mermaidchart.live
See in Classroom

Updates on B+-Trees: Deletion

- **Case 2 :** The key to be deleted is present only at the leaf node not in the indexes (or internal nodes) and there is more than the minimum number of keys in the node(***no underflow***). Simply delete the key.

Deleting 40 from B+ tree (B+ of order 3)

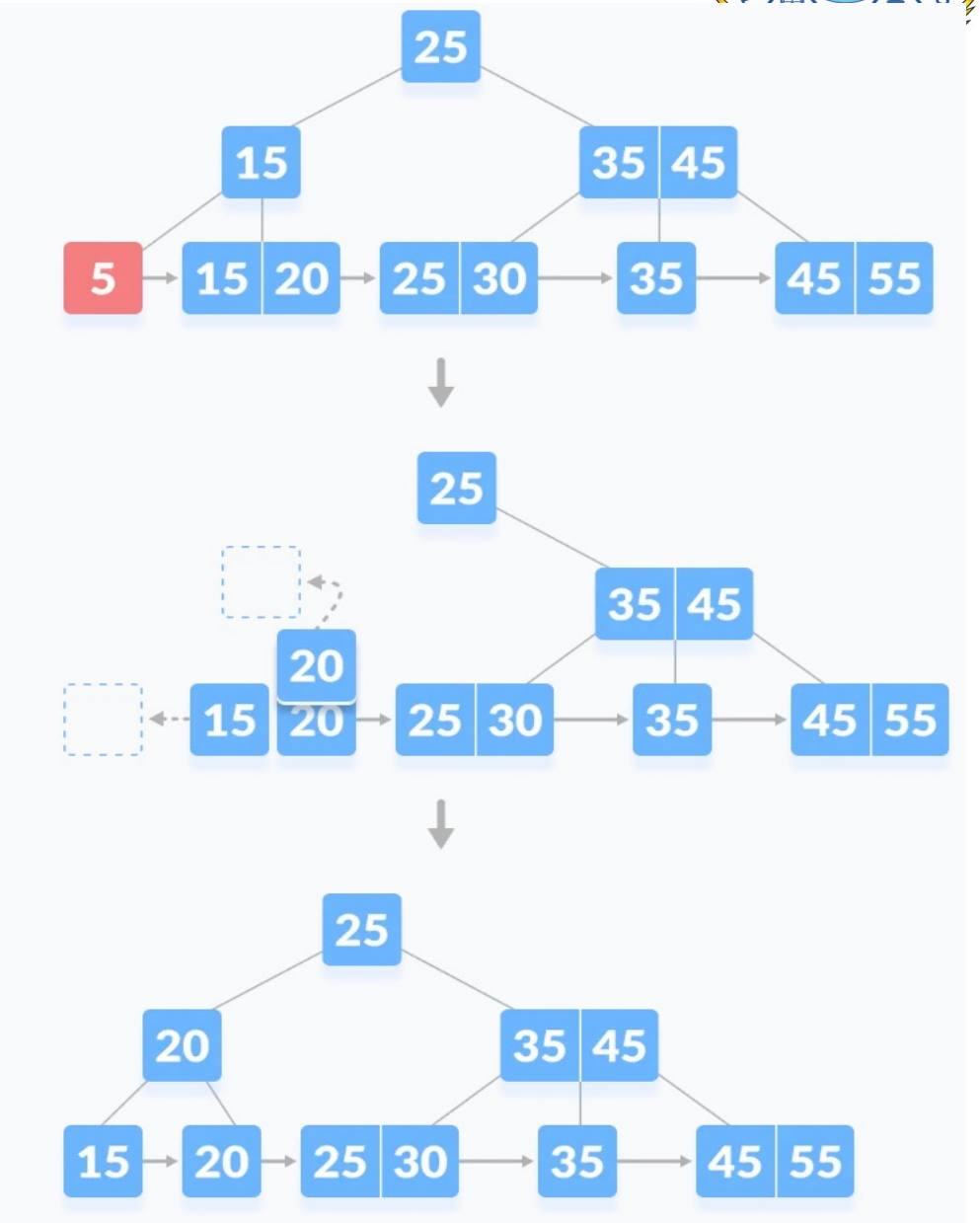


Updates on B+-Trees: Deletion

Case 3: Key in leaf node and there is an exact minimum number of keys in the node (***Cause Underflow and borrow possible***).

- Delete the key and borrow a key from the immediate sibling. Add the median key of the sibling node to the parent.

Delete 5 from B+ tree (B+ of order 3)

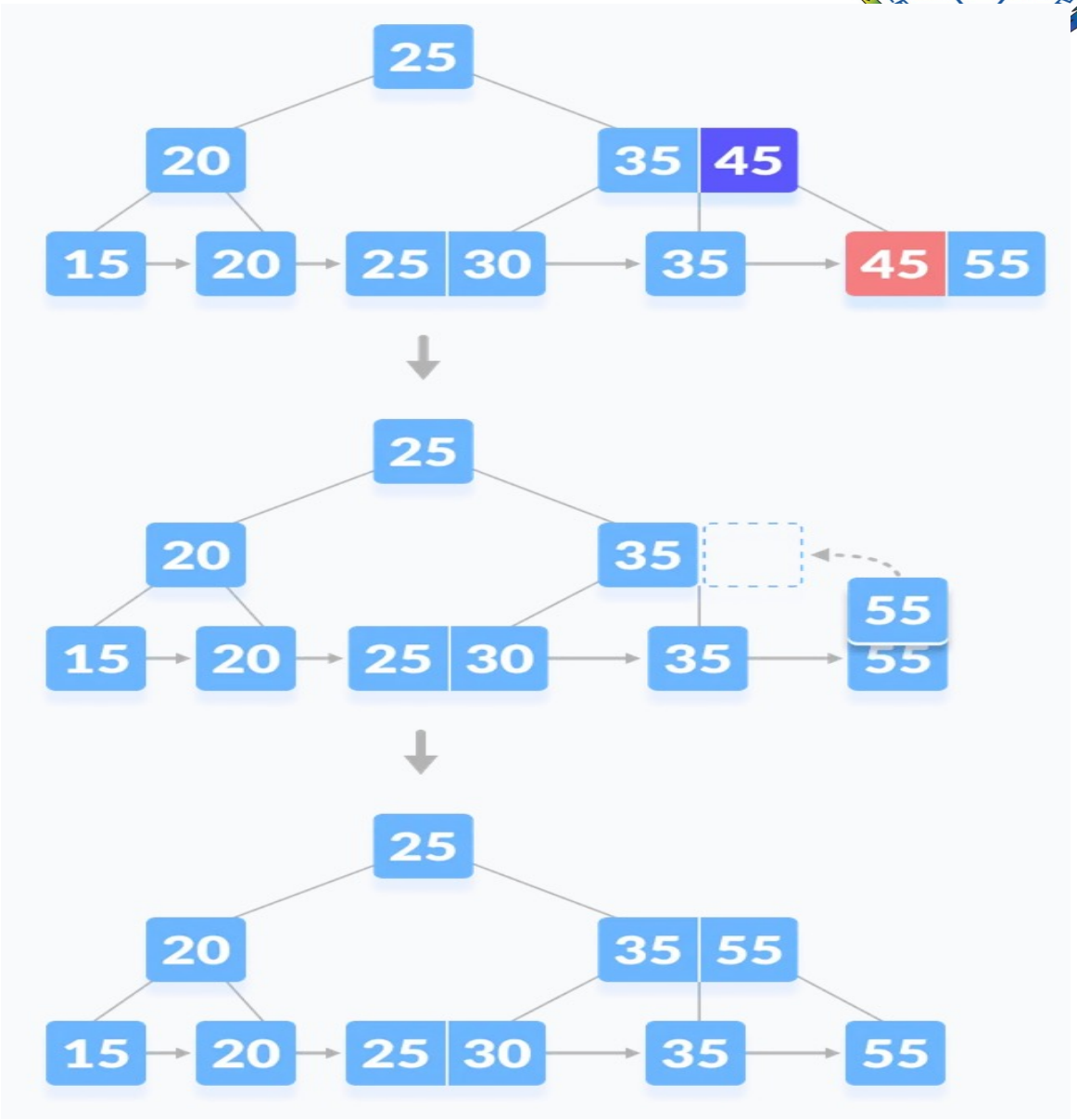


Updates on B+-Trees: Deletion

Case 5a: The key to be deleted is present in the internal nodes as well and there are more than minimum number of keys. Then remove the key from the internal nodes as well.

1. If there is more than the minimum number of keys in the node, simply delete the key from the leaf node and delete the key from the internal node as well. Fill the empty space in the internal node with the inorder successor.

- Delete 45 from B+ Tree

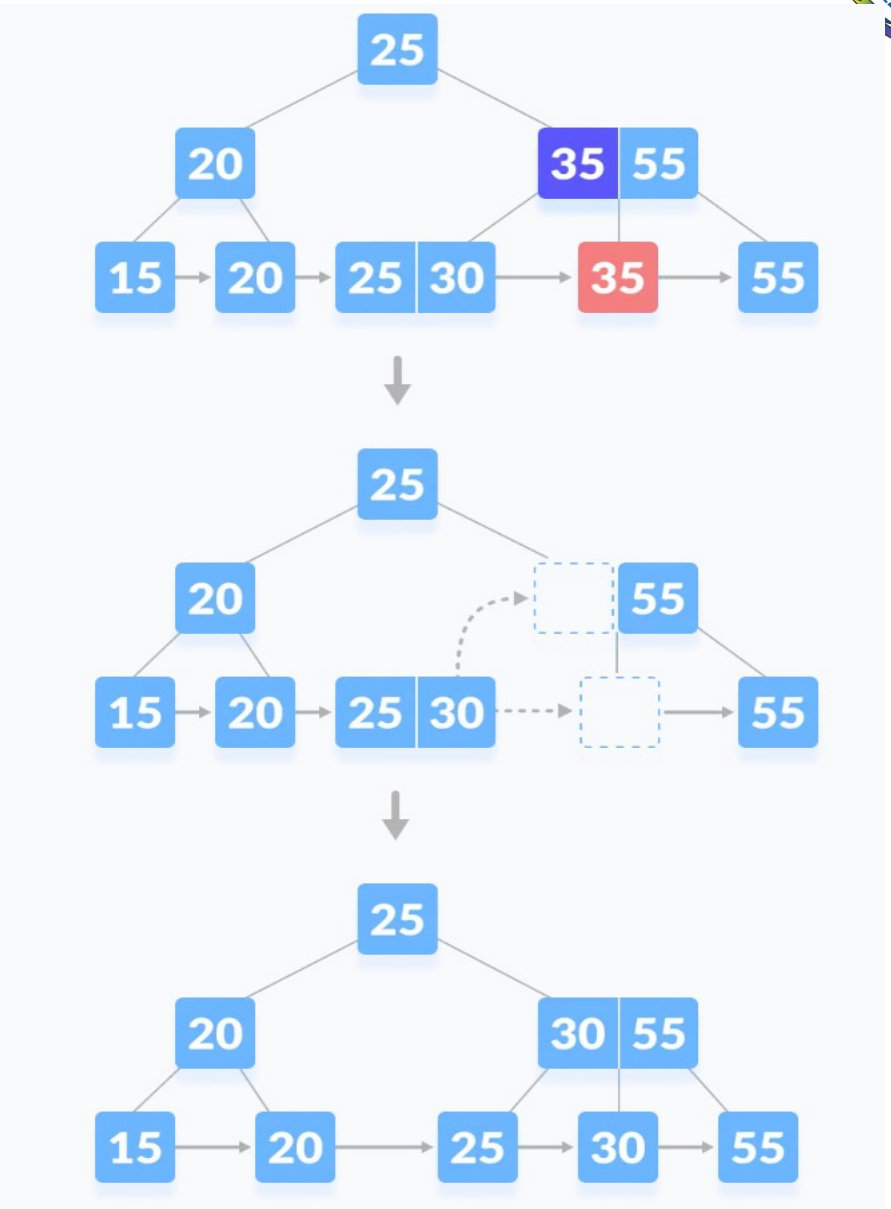


Updates on B+-Trees: Deletion

Case 5b: The key to be deleted is present in the internal nodes as well and there is an exact minimum number of keys in the node.

Then delete the key and borrow a key from its immediate sibling (through the parent).
Fill the empty space created in the index (internal node) with the borrowed key.

Delete 35 from B+ tree

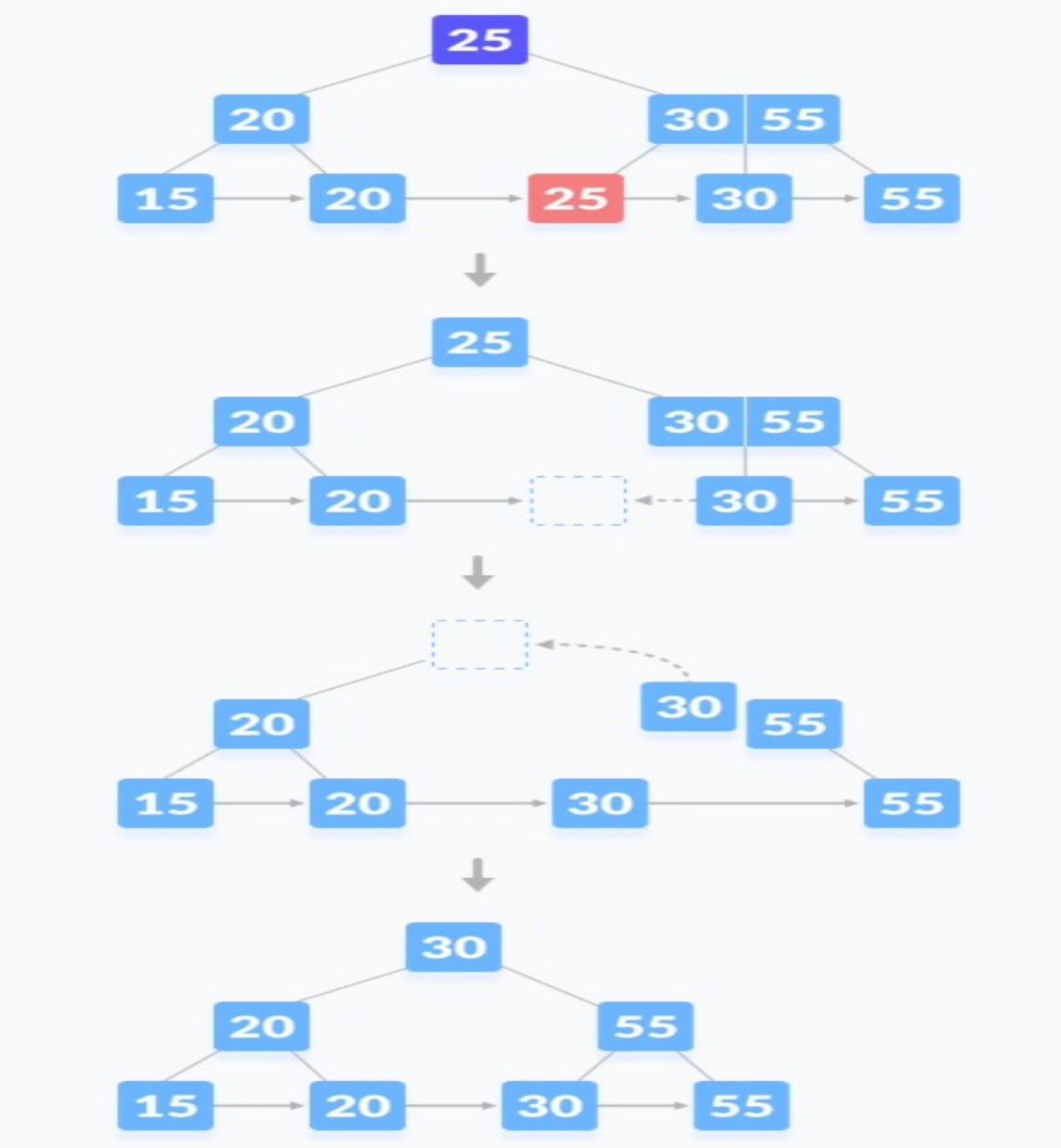


Updates on B+-Trees: Deletion

Case 5c: Key is present in leaf and internal node and there is exact minimum no of keys but an empty space is created above the immediate parent node.

- After deleting the key, merge the empty space with its sibling.
- Fill the empty space in the grandparent node with the inorder successor.

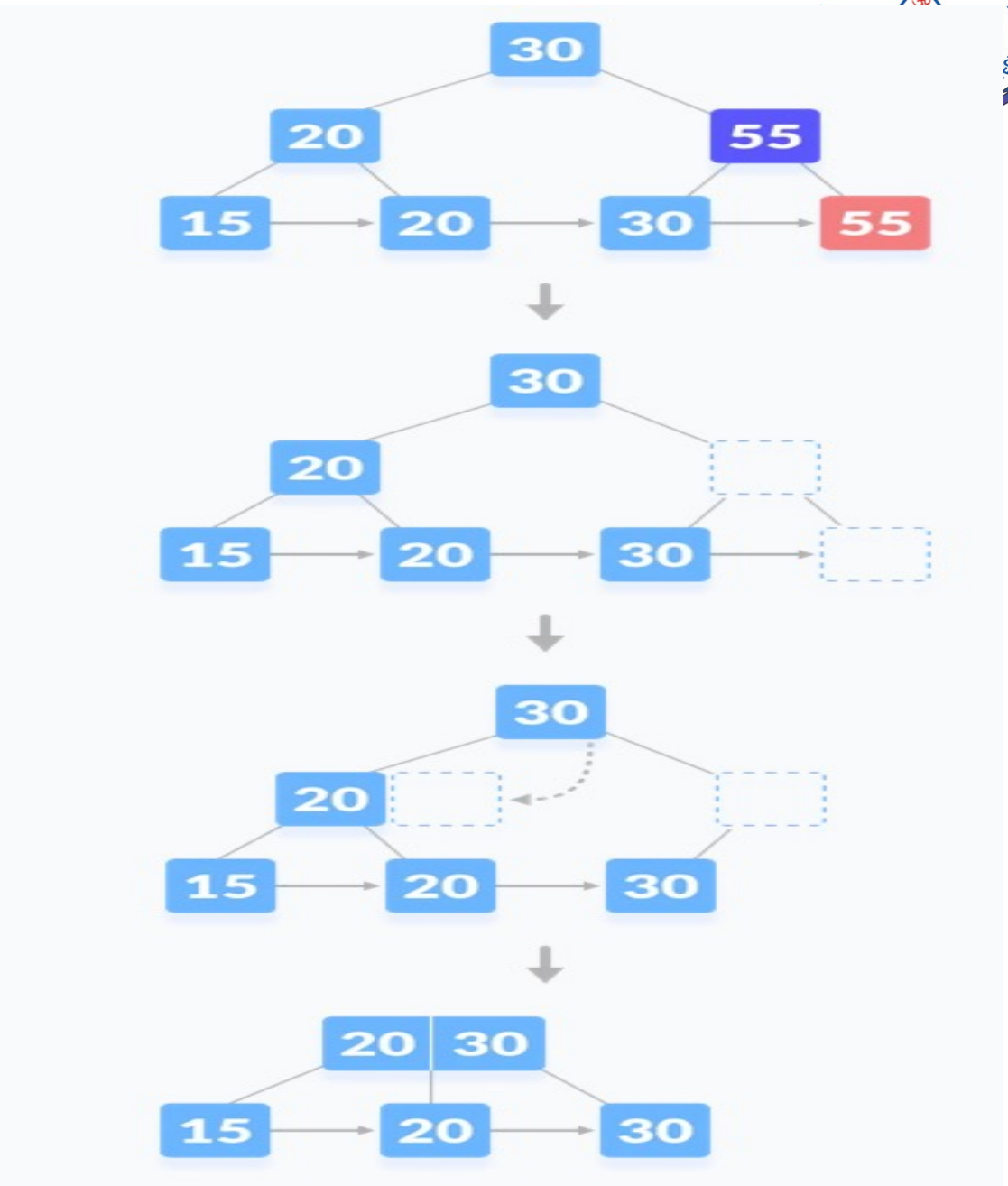
Delete 25 from B+ tree



Updates on B+-Trees: Deletion

Case 6: In this case, the height of the tree gets shrinked. It is a little complicated. Deleting 55 from the tree below leads to this condition.

Delete 55 from B+ tree



Complexity of Updates

- Cost (in terms of number of I/O operations) of insertion and deletion of a single entry proportional to height of the tree
  - With K entries and maximum fanout(**maximum number of child pointers)** of m, worst case complexity of insert/delete of an entry is $\mathbf{O(\log_{\lceil m/2 \rceil}(K))}$
- In practice, number of I/O operations is less:
  - Internal nodes tend to be in buffer
  - Splits/merges are rare, most insert/delete operations only affect a leaf node
- Average node occupancy depends on insertion order
  - 2/3rds with random, ½ with insertion in sorted order

Basis of Comparison	B tree	B+ tree
Pointers	All internal and leaf nodes have data pointers	Only leaf nodes have data pointers
Search	Since all keys are not available at leaf, search often takes more time.	All keys are at leaf nodes, hence search is faster and more accurate.
Redundant Keys	No duplicate of keys is maintained in the tree.	Duplicate of keys are maintained and all nodes are present at the leaf.
Insertion	Insertion takes more time and it is not predictable sometimes.	Insertion is easier and the results are always the same.
Deletion	Deletion of the internal node is very complex and the tree has to undergo a lot of transformations.	Deletion of any node is easy because all node are found at leaf.
Leaf Nodes	Leaf nodes are not stored as structural linked list.	Leaf nodes are stored as structural linked list.
Access	Sequential access to nodes is not possible	Sequential access is possible just like linked list
Height	For a particular number nodes height is larger	Height is lesser than B tree for the same number of nodes
Application	B-Trees used in Databases, Search engines	B+ Trees used in Multilevel Indexing, Database indexing
Number of Nodes	Number of nodes at any intermediary level 'l' is 2^l.	Each intermediary node can have n/2 to n children.

Complexity Comparison of B tree and B+ tree

Operation	B-Tree Complexity	B+ Tree Complexity
Search	$O(\log_m K)$	$O(\log_m K)$
Insertion	$O(\log_m K)$	$O(\log_m K)$
Deletion	$O(\log_m K)$	$O(\log_m K)$
Range Query	$O(\log_m K+t)$	$O(\log_m K+t)$
Height	$O(\log_m K)$	$O(\log_m K)$

K: Number of keys in the tree.

m: Fanout (maximum number of children per node).

t: Number of keys returned in a range query.

When to Use B-tree vs B+ tree

Use B-tree:

1. When **space utilization** is critical (less redundancy).
2. For smaller datasets where range queries are infrequent.
3. When you need to minimize pointer overhead since data can exist in internal nodes.

Use B+ tree:

1. For **databases** and **file systems** where range queries and sequential access are common.
2. When disk I/O optimization is crucial due to linked leaf nodes.
3. For very large datasets, as search time and space are optimized.

Outline

1. Advanced Trees
   - B and B+ Trees
2. Advanced Graph
   - Planar Graphs and its applications in Geographic and Circuit Layout Design
3. Network Flow Algorithms
   - Ford-Fulkerson Algorithm
   - Edmonds-Karp Algorithm
4. Advanced Shortest Path Algorithms
   - Bellman-Ford Algorithm
   - Johnson's algorithm
5. Graphical Data Structures
   - Quadtrees, KD Trees, R-Trees
6. Computational Geometry
   - Convex Hull and Voronoi Diagrams
7. Case Studies in Red-Black Tree and Its Applications, Applications of Directed Acyclic Graph, Applications of Minimum Spanning Tree in Network Design

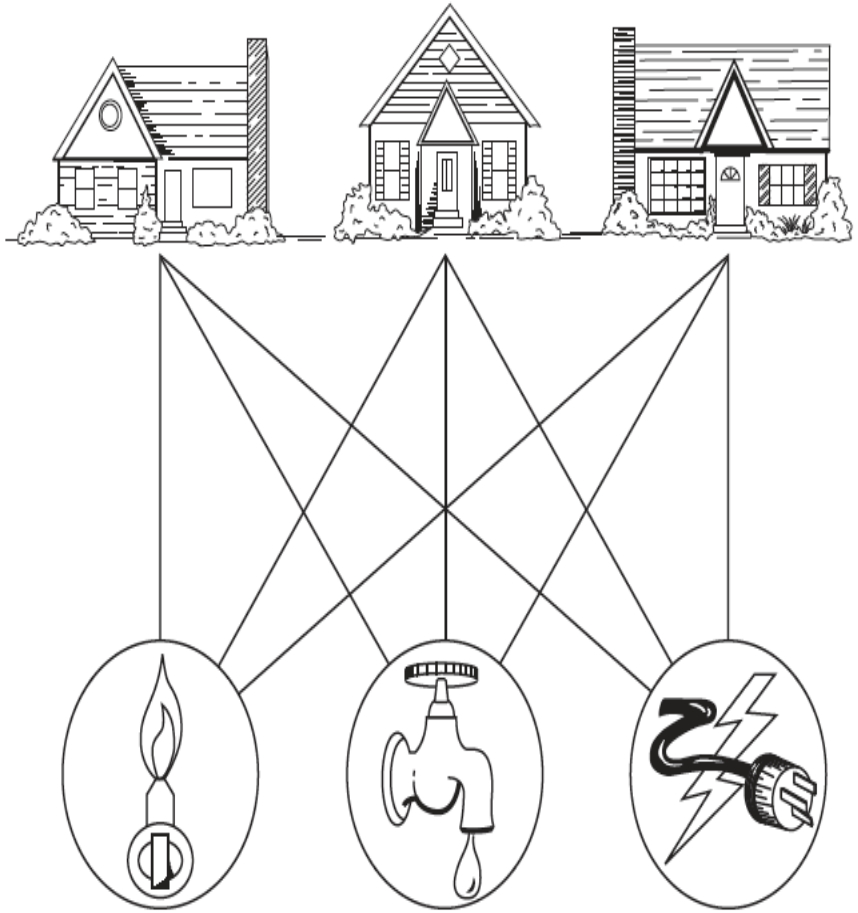
Advanced Graph

Advanced Graph: Planar Graphs and its applications in Geographic and Circuit Layout Design

Planar Graphs

- Consider the problem of joining three houses to each of three separate utilities as in Figure. Is it possible to join these houses and utilities so that none of the connections cross?

- There are always many ways to represent a graph. When is it possible to find at least one way to represent this graph in a plane without any edges crossing?



Planar Graphs

1. A graph is called **planar** if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a planar representation of the graph.
2. A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.
3. A planar representation of a graph splits the plane into regions, including an unbounded region.
4. A complete graph of five vertices is non planar.

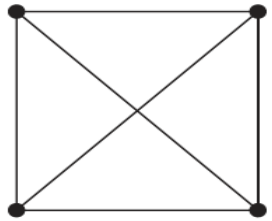


Fig: The Graph K4.

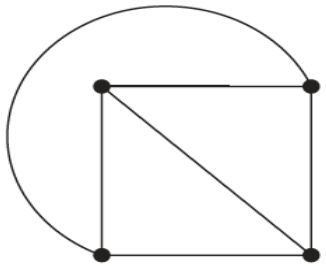


Fig: K4 Drawn with No Crossings.

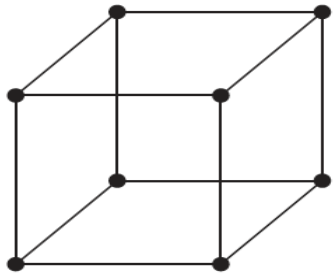


Fig: The Graph Q3.

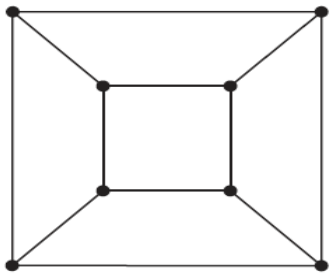
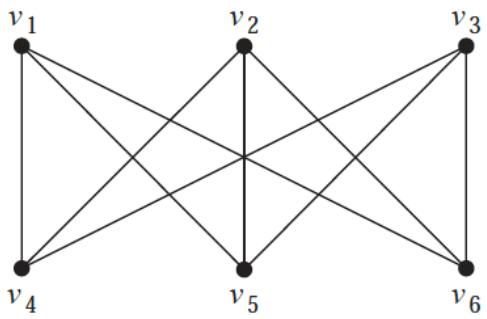
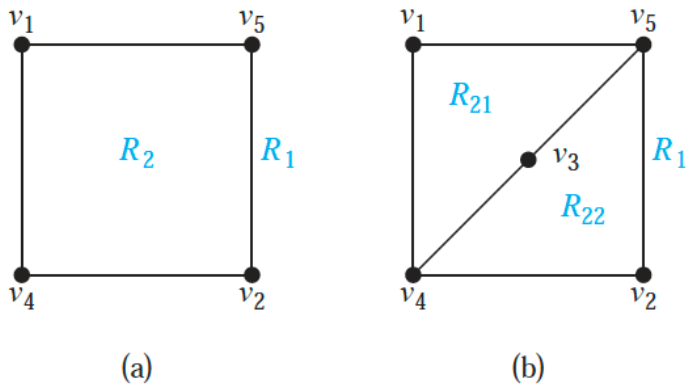


Fig: A Planar Representation of Q3.

Is $K_{3,3}$ graph(in figure shown) is planar?





- Any attempt to draw $K_{3,3}$ in the plane with no edges crossing is doomed.
- In any planar representation of $K_{3,3}$, the vertices v_1 and v_2 must be connected to both v_4 and v_5. These four edges form a closed curve that splits the plane into two regions, R_1 and R_2, as shown in Figure (a). The vertex v_3 is in either R_1 or R_2. When v_3 is in R_2, the inside of the closed curve, the edges between v_3 and v_4 and between v_3 and v_5 separate R_2 into two subregions, R_{21} and R_{22}, as shown in Figure (b).

Planar Graphs

Applications of Planar Graph

- Planarity of graphs plays an important role in the design of electronic circuits. We can model a circuit with a graph by representing components of the circuit by vertices and connections between them by edges. We can print a circuit on a single board with no connections crossing if the graph representing the circuit is planar. When this graph is not planar, we must turn to more expensive options.
- For example, we can partition the vertices in the graph representing the circuit into planar subgraphs. We then construct the circuit using multiple layers.
- We can construct the circuit using insulated wires whenever connections cross. In this case, drawing the graph with the fewest possible crossings is important.
- The planarity of graphs is also useful in the design of road networks.
- Suppose we want to connect a group of cities by roads. We can model a road network connecting these cities using a simple graph with vertices representing the cities and edges representing the highways connecting them.
- We can built this road network without using underpasses or overpasses if the resulting graph is planar.

Assignment

Do some research and write a synopsis paper including case studies where planar graph can be use for Circuit layout design and GIS. Submit it in Google classroom.